

Measurement-Based Quantum Diffusion Models

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We introduce measurement-based quantum diffusion models that bridge classical and quantum diffusion theory through randomized weak measurements. The measurement-based approach naturally generates stochastic quantum trajectories while preserving purity at the trajectory level and inducing depolarization at the ensemble level. We address two quantum state generation problems: trajectory-level recovery of pure state ensembles and ensemble-average recovery of mixed states. For trajectory-level recovery, we establish that quantum score matching is mathematically equivalent to learning unitary generators for the reverse process. For ensemble-average recovery, we introduce local Petz recovery maps for states with finite correlation length and classical shadow reconstruction for general states, both with rigorous error bounds. Our framework establishes Petz recovery maps as quantum generalizations of reverse Fokker-Planck equations, providing a rigorous bridge between quantum recovery channels and classical stochastic reversals. This work enables new approaches to quantum state generation with potential applications in quantum information science.

I. INTRODUCTION

Diffusion-based generative models have achieving remarkable success in generating high-quality images, text, and other classical data structures through noise injection and denoising [1–3]. This success has naturally motivated the development of quantum analogs—quantum diffusion models that aim to generate quantum states rather than classical data. Recent years have witnessed growing interest in this direction, with several frameworks proposed [4–14]. These approaches have demonstrated the ability to generate both pure and mixed quantum states with high fidelity, unlock a range of applications in quantum information science—from state preparation to quantum error correction.

However, despite these advances, a fundamental gap remains between classical and quantum diffusion models. Classical diffusion theory is built upon a rich mathematical framework [15] (see Tab. I) that connects stochastic differential equations (SDEs), ordinary differential equations (ODEs) and partial differential equations (PDEs) through the *score function*—the gradient of log probability. This framework provides not only a unified understanding of forward and reverse processes but also principled training objectives through score matching. In contrast, existing quantum diffusion models lack this theoretical foundation: the role of score functions is unclear, the relationship between quantum SDEs and PDEs is not established, and training objectives are often constructed heuristically rather than derived from first principles.

In this work, we bridge this theoretical gap by establishing a complete correspondence between classical and quantum diffusion models. Our key insight is that quantum measurement naturally provides the stochastic processes needed for diffusion. We propose using *randomized weak measurements* as the forward diffusion process, which naturally gives rise to stochastic differential equations for quantum state evolution. This approach not

Classical Diffusion	Quantum Diffusion
SDE: $d\mathbf{x} = \mathbf{f} dt + g d\mathbf{w}$	$d \psi\rangle = (-\frac{\gamma}{2}\delta O^2 dt + \sqrt{\gamma}\delta O d\mathbf{w}) \psi\rangle$
ODE: $d\mathbf{x} = (\mathbf{f} - \frac{g^2}{2}\mathbf{s})dt$	$d \psi\rangle = -iH(\psi) \psi\rangle dt$
PDE: $\partial_t p = -\nabla \cdot ((\mathbf{f} - \frac{g^2}{2}\mathbf{s})p)$	$\partial_t \bar{\rho} = -\frac{\gamma}{2}[O, [O, \bar{\rho}]]$

TABLE I. Comparison between classical and quantum formulations of diffusion. Classical diffusion is driven by stochastic noise, while quantum diffusion is driven by repeated measurement of a random observable O with measurement strength γ . In classical diffusion, $p = p(\mathbf{x})$ denotes the probability distribution of classical random variable $\mathbf{x}$, and $\mathbf{s}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$ is the score function. In quantum diffusion, $\bar{\rho} = \int d\psi p(\psi) |\psi\rangle \langle \psi|$ characterizes the average state (density matrix), $\delta O = O - \langle \psi | O | \psi \rangle$ and $H(\psi)$ is a state-dependent control Hamiltonian.

only provides a physically realizable implementation but also establishes the missing theoretical connections.

We address two distinct but related problems in quantum state generation: the recovery of pure state ensembles and the recovery of average states (see Fig. 1). For pure state ensembles, we demonstrate that score matching in the quantum setting is mathematically equivalent to learning unitary generators (control Hamiltonian) for the reverse process. This equivalence provides a principled training objective that was previously missing in quantum diffusion literature. For average state recovery, we introduce two complementary approaches: local Petz recovery maps for states with finite correlation length and classical shadow reconstruction for general states. Both approaches come with rigorous error bounds that scale favorably with system size.

A key advantage of our measurement-based approach is that measurement serves dual purposes: it drives the forward stochastic process while simultaneously extracting information about the quantum state. This infor-

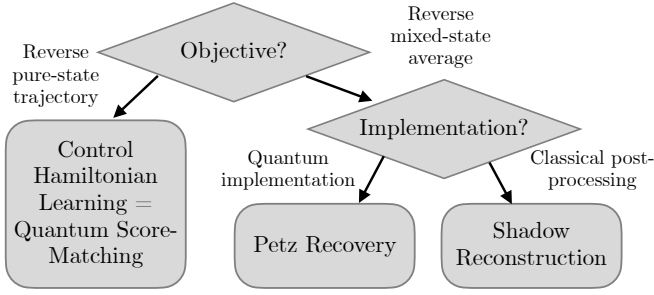


FIG. 1. Road map for reverse quantum diffusion.

mation can be used to condition the reverse process, enabling the construction of state-dependent control Hamiltonian or quantum processes. This capability is unique to measurement-based approaches and is not available in other quantum diffusion frameworks that rely on noise channels or unitary scrambling. The ability to condition on measurement outcomes enables deterministic, flow-based (unitary) reversal at the level of individual pure states, rather than, as in most existing literature [16], merely reconstructing the density matrix in an ensemble-averaged sense.

Our theoretical framework establishes several important connections. We show that the average behavior of the measurement-induced SDE is described by a Lindblad master equation, providing the quantum analog of the Fokker-Planck equation in classical diffusion. We prove that Petz recovery maps serve as the quantum generalization of reverse Fokker-Planck equations.

II. FORWARD QUANTUM DIFFUSION

A. Randomized Weak Measurements

Consider an n -qubit system initially prepared in a pure state $|\psi_0\rangle$. Under weak, continuously monitored measurements, the *conditional* state $|\psi_t\rangle$ remains pure at all times. Randomness of the measurement outcomes induces a *stochastic* trajectory of pure states (a quantum trajectory), given by a nonlinear stochastic differential equation (SDE): [17]

$$d|\psi_t\rangle = \left(-\frac{\gamma}{2}\delta O_t^2 dt + \sqrt{\gamma}\delta O_t dw_t \right) |\psi_t\rangle, \quad (1)$$

where $\delta O_t := O_t - \langle\psi_t| O_t |\psi_t\rangle$ is $|\psi_t\rangle$ -dependent, with O_t being the observable to be measured at time t . $\gamma > 0$ parameterize the *measurement strength*. dw_t describes a standard Wiener process satisfying $\mathbb{E}[dw_t] = 0$ and $\mathbb{E}[dw_t^2] = dt$, which captures the stochastic fluctuations of the *weak measurement outcome* do_t around its expectation value:

$$do_t = \langle\psi_t| O_t |\psi_t\rangle dt + \frac{dw_t}{2\sqrt{\gamma}}. \quad (2)$$

Operationally, do_t denotes the rescaled signal accumulated from weak measurements of O_t over the interval dt . For a single weak measurement within the dt interval, the observed eigenvalue of O_t corresponds to $2\sqrt{\gamma/dt} do_t$.

In classical diffusion models, the forward process is typically designed as a trivializing transformation, mapping a complex target distribution to a simple reference distribution from which sampling is straightforward. To construct an analogous trivializing map for quantum diffusion, we consider a *randomized measurement* protocol, in which the observable O_t is drawn at each time step from a stochastic process independent of the evolving state $|\psi_t\rangle$. Randomized measurements generate an ensemble of pure states

$$\mathcal{E}_\psi = \{|\psi\rangle\langle\psi| : \psi \sim p(\psi)\}, \quad (3)$$

which can be either (i) described by the probability distribution $p(\psi)$ over the projective Hilbert space, or (ii) characterized by its first moment—the ensemble-averaged density matrix

$$\bar{\rho} = \mathbb{E}_{\psi \sim p(\psi)} |\psi\rangle\langle\psi|. \quad (4)$$

It is possible to choose the distribution of O_t such that, under randomized measurements, any initial ensemble evolves toward a uniform distribution over the Hilbert space. In this limit, the ensemble-averaged density matrix approaches the maximally mixed state, $\mathbb{E}_{\text{traj.}} |\psi_t\rangle\langle\psi_t| \xrightarrow{t \rightarrow \infty} \mathbb{1}/2^n$. This does not contradict the fact that each conditional state $|\psi_t\rangle$ along a single trajectory remains pure; the mixing arises solely from averaging over the classical randomness of the measurement record. Thus, randomized weak measurement can preserve purity at the trajectory level while inducing complete depolarization at the ensemble level, realizing a trivializing forward quantum diffusion.

Consequently, reverse quantum diffusion admits two natural generative objectives:

- (i) **Trajectory-level recovery:** sample from or reconstruct the trajectory-level pure-state distribution $p(\psi_t)$ induced by the forward process;
- (ii) **Ensemble-average recovery:** reconstruct the averaged state $\bar{\rho}_t = \mathbb{E}_{\psi_t} |\psi_t\rangle\langle\psi_t|$ at each diffusion time t .

While most existing work [16] focuses on (ii), i.e., reconstructing $\bar{\rho}_t$, the trajectory-level task (i) is considerably more challenging, as it requires a classical decoder to infer the underlying pure state from the stochastic measurement record of each trajectory.

In this work, we address both objectives, as illustrated in Fig. 1. For trajectory-level recovery (i), we develop a *control Hamiltonian learning* approach that learns a time-reversal unitary evolution from measurement records. For ensemble-average recovery (ii), we consider two complementary strategies: a *Petz recovery* map for quantum implementation, and *classical shadow reconstruction* for purely classical post-processing.

B. Measurement-Based Forward Diffusion

According to Eq. (1), the state projection operator $\rho_t = |\psi_t\rangle\langle\psi_t|$ (pure-state density matrix) evolves along the quantum trajectory as: [18]

$$d\rho_t = -\frac{\gamma}{2}[O_t, [O_t, \rho_t]] dt + \sqrt{\gamma}\{\delta O_t, \rho_t\} dw_t. \quad (5)$$

The first term in Eq. (5) represents a deterministic drift towards eigenbasis of O_t , arising from the measurement-induced decoherence. The second term captures the stochastic back-action generated by the quantum state collapse under measurement.

Let $\mathbf{P} = \{P_i\}_{i=1}^{4^n}$ denote the n -qubit Pauli-operator basis with $\text{Tr}(P_i P_j) = 2^n \delta_{ij}$. Any state ρ_t or observable O_t can be expanded as

$$\rho_t = \frac{\mathbf{z}_t \cdot \mathbf{P}}{2^n}, \quad O_t = \mathbf{x}_t \cdot \mathbf{P}, \quad (6)$$

with $z_{t,i} = \text{Tr}(\rho_t P_i) = \langle\psi_t|P_i|\psi_t\rangle$, $x_{t,i} = \text{Tr}(O_t P_i)/2^n$. Then Eq. (5) implies

$$d\mathbf{z}_t = \mathbf{f}(\mathbf{z}_t, t) dt + \mathbf{g}(\mathbf{z}_t, t) dw_t, \quad (7)$$

with the drift $\mathbf{f}$ and the noise $\mathbf{g}$ functions given by (Einstein summation convention implied)

$$\begin{aligned} f_l(\mathbf{z}_t, t) &= -\gamma(c_{ijm}c_{klm} - c_{ikm}c_{jlm})x_{t,i}x_{t,j}z_{t,k}, \\ g_l(\mathbf{z}_t, t) &= \sqrt{\gamma}((c_{ijl} + c_{jil})x_{t,i}z_{t,j} - 2x_{t,i}z_{t,i}z_{t,l}), \end{aligned} \quad (8)$$

and $c_{ijk} = \text{Tr}(P_i P_j P_k)/\text{Tr} \mathbf{1}$ being the operator product expansion (OPE) coefficient.

The SDE Eq. (7) for $\mathbf{z}_t$ takes the form of a classical Langevin equation, linking measurement-based quantum diffusion directly to classical diffusion. Here, the Pauli expectation values $\mathbf{z}_t$ serve as stochastic variables encoding the quantum state ρ_t . If accessible, the time series $\{\mathbf{z}_t\}$ could train a denoising model for reversal. However, in quantum systems, $\mathbf{z}_t$ (or ρ_t) is not directly observable, and must be inferred from the measurement record $\{(O_t, o_t)\}$. We will come back to this problem later in Sec. III A 1.

C. Kraus Operator Formulation

The stochastic evolution of $|\psi_t\rangle$ in Eq. (1) can be equivalently expressed in the Kraus operator formalism:

$$|\psi_{t+dt}\rangle = \frac{K_{dt}(O_t, do_t) |\psi_t\rangle}{\|K_{dt}(O_t, do_t) |\psi_t\rangle\|}, \quad (9)$$

where K_{dt} denotes the *Kraus operator* for a weak measurement of observing O_t and obtaining the measurement outcome signal do_t :

$$K_{dt}(O_t, do_t) = \left(\frac{2\gamma}{\pi dt}\right)^{\frac{1}{4}} \exp\left(-\frac{\gamma}{dt} (do_t - O_t dt)^2\right). \quad (10)$$

The probability density for observing do_t is given by $p(do_t|O_t, \psi_t) = \|K_{dt}(O_t, do_t) |\psi_t\rangle\|^2$, which follows a Gaussian distribution centered at $\langle\psi_t|O_t|\psi_t\rangle dt$ with variance $(4\gamma)^{-1}dt$, consistent with Eq. (2).

The randomized measurement not only drives the diffusion of the conditional pure state $|\psi_t\rangle$, but also generates a *weak measurement record* along the trajectory, represented by the time series

$$\mathcal{O} = \{(O_t, o_t) \mid 0 \leq t \leq T\}. \quad (11)$$

The accumulated Kraus operator along the measurement trajectory $\mathcal{O}$ is

$$K_t(\mathcal{O}) = \mathcal{T} \prod_{t'=0}^{t-dt} K_{dt}(O_{t'}, do_{t'}), \quad (12)$$

where $\mathcal{T}$ denotes time-ordering.

D. Pauli Twirled Measurement Channel

Averaging over all possible measurement trajectories defines the weak measurement channel $\mathcal{F}_t$:

$$\bar{\rho}_t = \mathcal{F}_t(\bar{\rho}_0) = \mathbb{E}_{\mathcal{O}} K_t(\mathcal{O}) \bar{\rho}_0 K_t^\dagger(\mathcal{O}), \quad (13)$$

which describes the ensemble-averaged evolution under the forward diffusion process. Physically, $\mathcal{F}_t$ captures the decoherence induced by weak measurements after averaging over the classical randomness of observable choices O_t and measurement outcomes o_t in $\mathcal{O} = \{(O_t, o_t)\}$.

Averaging the stochastic evolution in Eq. (5) over all measurement trajectories transforms the SDE for the pure-state projector ρ_t into a deterministic PDE for the average state $\bar{\rho}_t$. The resulting evolution is governed by a Lindblad master equation,

$$\partial_t \bar{\rho}_t = \mathcal{L}[\bar{\rho}_t] = -\frac{\gamma}{2} \mathbb{E}_{O_t} [O_t, [O_t, \bar{\rho}_t]], \quad (14)$$

where the specific form of the Liouvillian super-operator $\mathcal{L}$ depends by the observable distribution $p(O_t)$. Equivalently, the channel can be written as $\mathcal{F}_t = e^{t\mathcal{L}}$.

The measurement channel $\mathcal{F}_t$ is said to be *Pauli twirled* [19–22] if it is weakly symmetric [23–25] under local Clifford transformations, i.e. $\forall U \in C_1^{\otimes n}$ (C_1 denotes the single-qubit Clifford unitary group):

$$\mathcal{F}_t(U \bar{\rho}_0 U^\dagger) = \mathcal{F}_t(\bar{\rho}_0) = U \mathcal{F}_t(\bar{\rho}_0) U^\dagger. \quad (15)$$

Formally, Pauli twirled measurement channels can be constructed by requiring the observable distribution $p(O_t)$ to be invariant under local Clifford unitaries, i.e. $p(O_t) = p(U^\dagger O_t U)$. The nice property of Pauli twirled channels is that they are diagonal in the Pauli basis, such that given the decomposition $\bar{\rho}_t = \mathbf{z}_t \cdot \mathbf{P}/2^n$, the action of the channel $\bar{\rho}_t = \mathcal{F}_t(\bar{\rho}_0)$ simply amounts to rescaling each Pauli expectation value

$$z_{t,i} = w_{\mathcal{F}_t}(P_i) z_{0,i} \quad (16)$$

by the *Pauli weight* $w_{\mathcal{F}_t}(P) := \text{Tr}(P\mathcal{F}_t(P)) / \text{Tr} \mathbb{1}$ that characterizes the decay of each Pauli component under the channel.

E. Randomized Pauli Measurement

To enable concrete analysis and numerical simulation, we focus on a discrete-time weak-measurement scheme with small but finite time steps of duration $\delta t > 0$. At each step, the measured observable O_t is drawn uniformly at random from single-qubit Pauli operators

$$\mathcal{P}_1 = \{ \sigma_{x,j}, \sigma_{y,j}, \sigma_{z,j} \mid j = 1, \dots, n \}. \quad (17)$$

Unless otherwise stated, all subsequent results will refer to this single-qubit measurement setting.

As Pauli observables $O_t \in \mathcal{P}_1$ satisfy $O_t^2 = \mathbb{1}$ with eigenvalues ± 1 , the measurement signal $\delta o_t = \pm \frac{1}{2} \sqrt{\delta t} / \gamma$ also take binary values correspondingly. In this case, the differential Kraus operator in Eq. (10) reduces to the following discrete form after proper normalization,

$$K_{\delta t}(O_t, \delta o_t) = \frac{1}{\sqrt{2}} \exp\left(-\gamma \delta t + 2\gamma O_t \delta o_t\right). \quad (18)$$

Averaging over outcomes and the random choice of O_t yields the single-step measurement channel

$$\begin{aligned} \bar{\rho}_{t+\delta t} &= \mathcal{F}_{\delta t}(\bar{\rho}_t) \\ &= \mathbb{E}_{O_t, \delta o_t} K_{\delta t}(O_t, \delta o_t) \bar{\rho}_t K_{\delta t}^\dagger(O_t, \delta o_t). \end{aligned} \quad (19)$$

This acts as a product of single-qubit depolarization channel acting identically and independently across all qubits. The corresponding finite-difference Lindblad equation reads $\delta \bar{\rho}_t = \mathcal{L}[\bar{\rho}_t] \delta t$, with the Liouvillian super-operator decomposes to each qubit $\mathcal{L} = \frac{1}{n} \sum_{j=1}^n \mathcal{L}_j$,

$$\mathcal{L}_j[\bar{\rho}_t] = -\frac{4\gamma}{3} \left(\bar{\rho}_t - (\text{Tr} \bar{\rho}_t) \frac{\mathbb{1}}{2} \right), \quad (20)$$

which precisely generates local depolarization.

Because the channel is Pauli-twirled, its action on the n -qubit Pauli basis is diagonal. The Pauli weight depends only on the *operator weight* $|P|$ (the number of qubits on which P acts nontrivially) and takes the closed form

$$w_{\mathcal{F}_t}(P) = \exp\left(-\frac{4\gamma}{3n} |P| t\right), \quad (21)$$

describing exponential decay of each nontrivial Pauli component at a uniform rate $4\gamma/3n$ per qubit.

III. REVERSE QUANTUM DIFFUSION

A. Control Hamiltonian Learning

1. Classical Decoder

The reverse diffusion task can be formulated as a control problem: given the current state of the system, a

denoising model must condition on this state and propose control operations that drive the system backward in time. In classical systems, the state variables coincide with physical observables, which can be directly accessed, copied, and processed by the denoising model to generate classical control actions. In quantum systems, however, there is a fundamental separation between the underlying quantum state and measurable observables. The quantum state cannot be directly accessed; it can only be inferred from repeated measurements. Yet, quantum trajectories are not reproducible, making it impossible to reconstruct intermediate states along the trajectory via quantum state tomography without severe post-selection overhead. As a result, a quantum denoising algorithm does not have direct access to the state and must base its control on indirect and noisy inference, which is substantially more challenging than in the classical case.

To address this difficulty, we introduce a *classical decoder* that works alongside the quantum denoising model. The decoder is a classical algorithm that infers the sequence of Pauli expectation values $\{\mathbf{z}_t\}$ from the measurement record $\{(O_t, o_t)\}$. Such a decoder can be implemented using a classical simulator or, alternatively, learned as a data-driven inference model. In this work, to demonstrate the feasibility of this approach, we employ a simulation-based decoder, acknowledging that it is not scalable to large quantum systems.

The Kraus-operator formulation in Sec. II C provides a concrete basis for constructing a classical decoder. Given a measurement record $\mathcal{O}$, the classical decoding algorithm proceeds as:

- (i) Infer the initial state

$$|\psi_0\rangle = \arg \max_{|\psi\rangle} \|K_T(\mathcal{O}) |\psi\rangle\|^2 \quad (22)$$

by maximum-likelihood estimation.

- (ii) Reconstruct the intermediate states by forward Kraus map

$$|\psi_t\rangle = \frac{K_t(\mathcal{O}) |\psi_0\rangle}{\|K_t(\mathcal{O}) |\psi_0\rangle\|}. \quad (23)$$

- (iii) Compute the Pauli expectation values $\mathbf{z}_t = \langle \psi_t | \mathbf{P} | \psi_t \rangle$ for all time t to construct $\{\mathbf{z}_t\}$. These calculations assume the availability of a classical simulator.

While this simulation-based method might not be the most efficient form of classical decoding, it serves as a proof-of-principle. More scalable alternatives, such as data-driven learning-based decoders, will be left for future investigation.

2. Quantum Score Matching

Given the time series $\{\mathbf{z}_t\}$ decoded from the weak measurement record, reversing the diffusion becomes a quan-

tum control problem. The objective is to learn a control Hamiltonian $H_\theta(\mathbf{z}_t, t)$ (with trainable parameter θ) that drives $|\psi_t\rangle$ backward along its trajectory. Since pure states remain pure under the forward diffusion Eq. (1), the time-reversed backward process must also preserve purity and can therefore be generated by a unitary evolution, described by the Schrödinger equation (as an ODE)

$$d|\psi_{t+dt}\rangle = i H_\theta(\mathbf{z}_{t+dt}, t) |\psi_{t+dt}\rangle dt. \quad (24)$$

This is analogous to the probability flow ODE approach in classical diffusion models [2].

We introduce a *unitary score function* $V_\theta(\mathbf{z}_{t+dt}, t) := \exp(i H_\theta(\mathbf{z}_{t+dt}, t) dt)$, representing the dt -time unitary operator that approximates the average backward flow from $|\psi_{t+dt}\rangle$ to $|\psi_t\rangle$. It is trained by minimizing the mean infidelity loss

$$\mathcal{L}_\theta = 1 - \mathbb{E}_{\psi_t, \psi_{t+dt}} |\langle \psi_t | V_\theta(\mathbf{z}_{t+dt}, t) | \psi_{t+dt} \rangle|^2, \quad (25)$$

where $|\psi_{t+dt}\rangle = K_{dt}(O_t, do_t) |\psi_t\rangle$ is generated by a step of forward diffusion, and $\mathbf{z}_{t+dt} = \langle \psi_{t+dt} | \mathbf{P} | \psi_{t+dt} \rangle$ is the corresponding Pauli expectation values. Both can be inferred from measurement data given a classical decoder/simulator. Once trained, the reverse process is implemented by applying $V_\theta(\mathbf{z}, t)$ at each step, generating a purity-preserving backward evolution.

This formulation is formally equivalent to the score-matching denoising loss for the SDE in Eq. (7):

$$\mathcal{L}_\theta = \frac{1}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left\| \mathbf{s}_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t, t) dt, t) + \frac{\mathbf{z}_{t+dt} - \mathbf{z}_t - \mathbf{f}(\mathbf{z}_t, t) dt}{dt} \right\|^2. \quad (26)$$

In classical diffusion models, the score function $\mathbf{s}(\mathbf{z}, t) = \nabla_{\mathbf{z}} \log p_t(\mathbf{z})$ is the gradient of the log-likelihood and determines the drift required to reverse the diffusion process. A key distinction in the quantum setting is that the reverse drift takes the form $\mathbf{s}_\theta(\mathbf{z}, t) \rightarrow \nabla_{\mathbf{z}}[\Sigma(\mathbf{z}, t) \log p_t(\mathbf{z})]$, where $\Sigma_{ij}(\mathbf{z}, t) = g_i(\mathbf{z}, t) g_j(\mathbf{z}, t)$ is a state-dependent noise covariance matrix. This difference arises because quantum noise originates from measurement back-action that depends on the conditional state, whereas in classical diffusion—particularly in machine learning applications—Gaussian noise is added independently of the state. Despite this modification, the modeling and training of score functions are well established [15], enabling these methods to be adapted to the quantum setting, with the time series $\{\mathbf{z}_t\}$ serving as the classical training data.

By identifying

$$\begin{aligned} 2^{-n} (\mathbf{s}_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t, t), t) - \mathbf{f}(\mathbf{z}_t, t)) \cdot \mathbf{P} \\ = -i [H_\theta(\mathbf{z}_{t+dt}, t), \rho_t], \end{aligned} \quad (27)$$

where $\rho_t = |\psi_t\rangle \langle \psi_t|$. Eq. (25) and Eq. (26) become proportional. Thus, learning the SDE score term $\mathbf{s}_\theta - \mathbf{f}$ is equivalent to learning the control Hamiltonian H_θ that generates the time-reversal unitary.

3. Error Bound Analysis

Given a time-dependent control Hamiltonian $H(t)$, the reverse evolution along each pure-state trajectory from $t = T$ back to $t = 0$ is implemented by the unitary map

$$V[H] = \mathcal{T}^\dagger \exp\left(i \int_T^0 H(\mathbf{z}_{t+dt}, t) dt\right), \quad (28)$$

where $\mathcal{T}^\dagger$ denotes reverse time-ordering. Acting with $V[H]$ on each final state $|\psi_T\rangle$ generates a *reverse-propagated ensemble*

$$\mathcal{E}_0[H] = \{V[H] |\psi_T\rangle : |\psi_T\rangle \in \mathcal{E}_T\}, \quad (29)$$

where $\mathcal{E}_T$ denotes the final state ensemble $t = T$. The corresponding *average state* at $t = 0$ is obtained by averaging over the ensemble $\mathcal{E}_0[H]$,

$$\bar{\rho}_0[H] = \mathbb{E}_{\psi_T} V[H] |\psi_T\rangle \langle \psi_T| V^\dagger[H]. \quad (30)$$

To derive an error bound for the entire pure-state ensemble $\mathcal{E}_0[H]$ produced by the backward unitary evolution, we employ the Wasserstein distance as a metric on the space of pure-state ensembles, since the Wasserstein distance compares full probability distributions rather than just their averages. Suppose the learned generative model (with control Hamiltonian H_θ) produces an ensemble $\mathcal{E}_0[H_\theta] = \{|\phi\rangle \langle \phi| : \phi \sim p_\theta(\phi)\}$ of pure states drawn from a distribution p_θ , and that the true ensemble of pure states is $\mathcal{E}_0[H_{\text{true}}] = \{|\psi\rangle \langle \psi| : \psi \sim p_{\text{true}}(\psi)\}$ for some distribution p_{true} . The Wasserstein-1 distance (with trace distance as the cost function) between p_θ and p_{true} is defined as [8]

$$\begin{aligned} W_1(p_\theta, p_{\text{true}}) \\ = \inf_{\pi \in \Pi(p_\theta, p_{\text{true}})} \int D(|\phi\rangle, |\psi\rangle) \pi(|\phi\rangle, |\psi\rangle) d\phi d\psi, \end{aligned} \quad (31)$$

where $\Pi(p_\theta, p_{\text{true}})$ denotes the set of all couplings (joint distributions) $\pi(\phi, \psi)$ having marginals $p_\theta(\phi)$ and $p_{\text{true}}(\psi)$. In other words, $\pi(\phi, \psi)$ satisfies $\int \pi(\phi, \psi) d\phi = p_{\text{true}}(\psi)$ and $\int \pi(\phi, \psi) d\psi = p_\theta(\phi)$. The cost function $D(|\phi\rangle, |\psi\rangle)$ is the trace distance between the pure states $|\phi\rangle$ and $|\psi\rangle$.

Theorem 1. Distribution Convergence Theorem. *Let p_{true} be the true initial-state distribution, and let p_θ be the distribution obtained by running the reverse diffusion process for time $T = \mathcal{O}(n)/\gamma$ under the learned control unitary V_θ . Assume V_θ satisfies $\mathbb{E}_{\psi_t, \psi_{t+dt}} \sqrt{1 - |\langle \psi_{t+dt} | V_\theta(\mathbf{z}_{t+dt}, t) | \psi_t \rangle|^2} \leq \epsilon dt$ and that it is Lipschitz continuous in the sense that $D(V_\theta^\dagger(\mathbf{z})|\alpha\rangle, V_\theta^\dagger(\mathbf{z}')|\alpha\rangle) \leq L_V dt D(|\psi_z\rangle, |\psi_{z'}\rangle)$ for any pure state $|\alpha\rangle$, $\mathbf{z}$ and $\mathbf{z}'$. Then by choosing a stopping time $T = \frac{C}{L_V} \ln(1/\epsilon)$ with $0 < C < 1$, the Wasserstein-1 distance between p_θ and p_{true} is bounded by*

$$W_1(p_\theta, p_{\text{true}}) \leq \frac{\sqrt{2}}{L_V} \epsilon^{1-C} + \epsilon^{-C} E_{\text{diff}}(T) \quad (32)$$

with $E_{\text{diff}}(T)$ decaying exponentially in T as $ne^{-\frac{4\gamma}{3n}T}$. If we choose a sufficiently large measurement strength γ such that $\frac{4\gamma}{3nL_v} \geq 1$, the leading-order behavior of the bound for small ϵ is dominated by the first term in Eq. (32), and $W_1(p_0, p_{\text{true}}) \rightarrow 0$ as $\epsilon \rightarrow 0$.

Proof sketch. The 1-Wasserstein distance $W_1(p_\theta, p_{\text{true}})$ between the learned and true distributions accumulates over time due to the small per-step discrepancy ϵdt between their reverse diffusion processes. At each infinitesimal time step, this deviation is amplified at most by the Lipschitz factor $e^{L_v dt}$. Meanwhile, $E_{\text{diff}}(T)$ quantifies how far the distribution obtained by forward diffusion up to time T is from a uniform random product-state distribution, and this deviation is exponentially small in T . Combining these results and choose $T = \frac{C}{L_v} \ln(1/\epsilon)$ with $0 < C < 1$ yields the bound stated in the theorem. The proof details can be found in SM [26]. This means that the distribution distance $W_1 \rightarrow 0$ when the training error ϵ is made arbitrarily tiny.

4. Numerical Demonstration

To demonstrate the effectiveness of control-Hamiltonian learning, we consider a single-qubit example, as illustrated in Fig. 2 and Fig. 3, along with two two-qubit examples shown in Fig. 4. In the latter case, we consider a pure-state ensemble near the two-qubit spin singlet $\psi_B = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$. Perturbations are introduced as $|\tilde{\psi}\rangle = |\psi_B\rangle + \sum_{j=0}^3 \alpha_j |j\rangle$, where $|j\rangle$ runs over the computational basis states and α_j are independent complex Gaussian variables with standard deviation 0.1. After normalization, the resulting states $|\psi_0\rangle$ form the ensemble $\{|\psi_0\rangle\}$ representing the distribution of initial states in parameter space. The details of the two-qubit Heisenberg model can be found in the caption of Fig. 4.

B. Classical Shadow Reconstruction

1. Problem Statement

We now pivot from trajectory-wise reversal to *ensemble-average* recovery. In this setting, the data are no longer a single measurement record $\mathcal{O} = \{(O_t, o_t)\}$ tied to one quantum trajectory, but a collection of independent and identically distributed records $\{\mathcal{O}^{(s)}\}_{s=1}^M$ generated by repeatedly performing randomized weak-measurement protocol on the unknown initial state $\bar{\rho}_0$. The learning task is to reconstruct the $\bar{\rho}_0$ given the measurement record dataset.

This is a problem of learning quantum states from classical data. When the objective is inference—rather than coherent state preparation on a quantum device—*classical shadow tomography* [27] provides an efficient solution: it associates to each measurement record

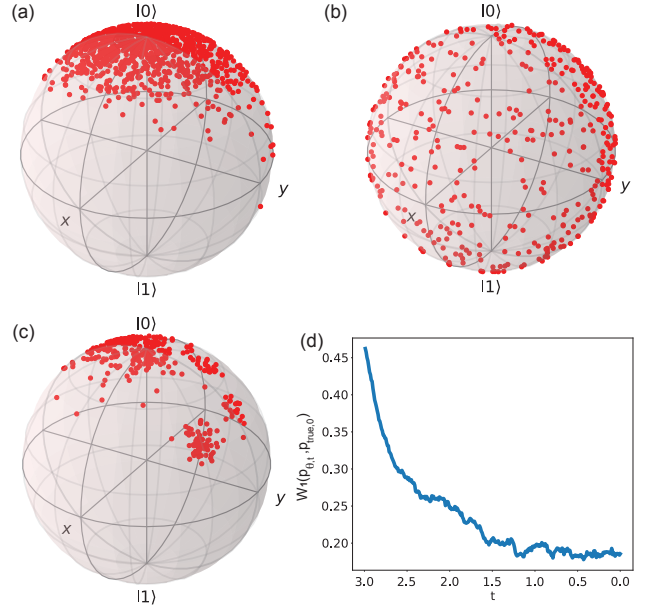


FIG. 2. Single qubit example with $T = 3, \gamma = 1, dt = 0.01$. (a) Training-state ensemble $\mathcal{E}_\psi = \{|\psi\rangle\langle\psi| : \psi \sim p(\psi)\}$ with $|\psi\rangle = |0\rangle + |\delta\psi\rangle$, where $|\delta\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $\alpha_{0,1}$ are complex Gaussian variables (standard deviation 0.1), normalized to unit norm. (b) Initial distribution of 400 samples for backward diffusion. (c) Final distribution of these samples after backward diffusion. (d) Decay of the Wasserstein distance $W(p_{\theta,t}, p_{\text{true},0})$ versus time t .

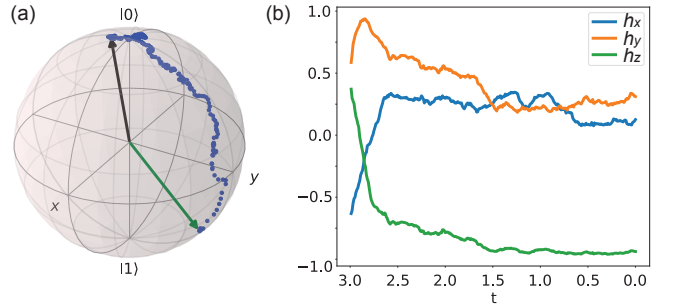


FIG. 3. Trajectory for the single qubit example in Fig. 2. (a) Single backward-diffusion trajectory. (b) Corresponding time evolution of the coefficients in $H_\theta = h_x\sigma_x + h_y\sigma_y + h_z\sigma_z$.

a classical shadow that is an unbiased randomized estimator of $\bar{\rho}_0$, and averaging classical shadows yields a reconstruction of $\bar{\rho}_0$ with rigorous concentration guarantees and favorable sample complexity.

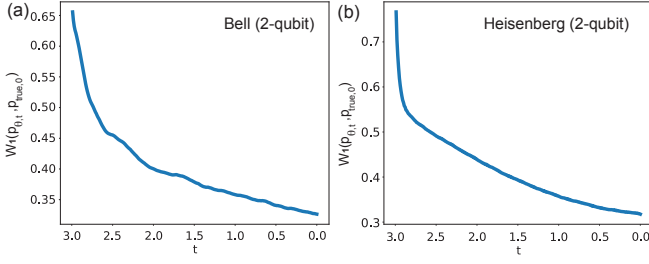


FIG. 4. Two qubit examples with $T = 3, \gamma = 1, dt = 0.01$. (a) $W_1(p_{\theta,t}, p_{\text{true},0})$ versus time t for a pure-state ensemble in the vicinity of the two-qubit spin singlet (Bell state) $|\psi_B\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$. (b) $W_1(p_{\theta,t}, p_{\text{true},0})$ versus time t for a thermal state ensemble of a two-qubit Heisenberg model $H = J\sigma_1 \cdot \sigma_2 + B_x\sigma_{1,x} + B_z\sigma_{2,x}$ with $B_x = B_z = 0.5$ and $J = 1$, in a temperature range $T_H \in [0, 0.5]$.

2. Measurement-and-Prepare Channel

For each measurement record $\mathcal{O}$ obtained from forward diffusion over time t , define the *measurement operator*

$$\sigma_t(\mathcal{O}) = K_t^\dagger(\mathcal{O}) K_t(\mathcal{O}) \quad (33)$$

in terms of the Kraus operator $K_t(\mathcal{O})$ defined in Eq. (12), such that the probability of observing $\mathcal{O}$ on an initial state $\bar{\rho}_0$ is

$$p(\mathcal{O} | \bar{\rho}_0) = \text{Tr}(\sigma_t(\mathcal{O}) \bar{\rho}_0), \quad (34)$$

referred to as the *posterior distribution* of $\mathcal{O}$.

Classical shadow tomography begins by repeatedly measuring identical copies of $\bar{\rho}_0$, thereby generating measurement records $\mathcal{O}$ drawn from the posterior distribution. Collecting M such measurement shots produces the dataset $\{\mathcal{O}^{(s)}\}_{s=1}^M$.

In the classical post-processing stage, each record $\mathcal{O}$ is mapped to its corresponding measurement operator $\sigma_t(\mathcal{O})$, regarded as a *classical snapshot*. This process can be described as a *measure-and-prepare channel*:

$$\begin{aligned} \mathcal{M}_t(\bar{\rho}_0) &= \sum_{\mathcal{O}} \sigma_t(\mathcal{O}) \text{Tr}[\sigma_t(\mathcal{O}) \bar{\rho}_0] \\ &= \mathbb{E}_{\mathcal{O} \sim p(\mathcal{O} | \bar{\rho}_0)} \sigma_t(\mathcal{O}). \end{aligned} \quad (35)$$

Here we intentionally do not normalize $\mathcal{M}_t$ (so it is not trace-preserving), as this choice simplifies the analysis.

The map $\mathcal{M}_t$ is a linear transformation from the target state $\bar{\rho}_0$ to the average classical snapshot. If the measurement scheme is *tomographically complete*—as in the randomized Pauli measurement case— $\mathcal{M}_t$ is invertible, and the state can be reconstructed as

$$\bar{\rho}_0 = \mathbb{E}_{\mathcal{O} \sim p(\mathcal{O} | \bar{\rho}_0)} \mathcal{M}_t^{-1}[\sigma_t(\mathcal{O})]. \quad (36)$$

Estimating the ensemble average requires samples from the quantum device, while the post-processing step constructs classical snapshots and applies $\mathcal{M}_t^{-1}$. This is the

basic principle of classical shadow reconstruction. The problem thus reduces to determining the inverse map $\mathcal{M}_t^{-1}$, also called the *reconstruction map*.

3. Channel Inversion and Pauli Weight

In general, computing $\mathcal{M}_t^{-1}$ is challenging, but our weak-measurement scheme allows an exact solution thanks to two key features:

- (i) *Pauli Twirling*. In randomized Pauli measurement, the observable distribution is invariant under local Clifford unitaries, implying that $\mathcal{M}_t$ is Pauli-twirled. Therefore, $\mathcal{M}_t$ is diagonal in the Pauli basis:

$$\mathcal{M}_t(P) = w_{\mathcal{M}_t}(P) P, \quad (37)$$

where the *Pauli weight* (as the channel eigenvalue) is defined as

$$w_{\mathcal{M}_t}(P) = \frac{\text{Tr}(P \mathcal{M}_t(P))}{\text{Tr} \mathbb{1}}. \quad (38)$$

- (ii) *Locality*. The randomized Pauli measurement is a local measurement scheme, meaning that weak measurements act independently on each qubit. This locality enables us to compute the Pauli weight for each qubit separately and combine the results multiplicatively. The final closed-form expression is

$$w_{\mathcal{M}_t}(P) = w_{\mathcal{M}_t}(\mathbb{1}) \left(\frac{1 - e^{-\frac{16\gamma t}{3n}}}{3 + e^{-\frac{16\gamma t}{3n}}} \right)^{|P|}, \quad (39)$$

where $|P|$ denotes the operator weight (support size) of the Pauli string P , and the factor

$$w_{\mathcal{M}_t}(\mathbb{1}) = \left(3e^{\frac{4\gamma t}{3n}} + e^{-\frac{4\gamma t}{n}} \right)^n \quad (40)$$

is a time-dependent function.

With these results, the reconstruction map is

$$\mathcal{M}_t^{-1}(P) = \frac{1}{w_{\mathcal{M}_t}(P)} P. \quad (41)$$

4. Estimating Observables and Sample Complexity

For finite sample size M , the Pauli expectation value can be estimated as

$$z_{0,i} \equiv \text{Tr}(\bar{\rho}_0 P_i) \approx \frac{1}{M} \sum_{s=1}^M \frac{\text{Tr}(\sigma_t(\mathcal{O}^{(s)}) P_i)}{w_{\mathcal{M}_t}(P_i)}, \quad (42)$$

from which the initial ensemble-average state $\bar{\rho}_0$ can be formally recovered as

$$\bar{\rho}_0 = \frac{\mathbf{z}_0 \cdot \mathbf{P}}{2^n} \approx \frac{1}{2^n M} \sum_{s=1}^M \frac{\text{Tr}(\sigma_t(\mathcal{O}^{(s)}) P_i)}{w_{\mathcal{M}_t}(P_i)} P_i, \quad (43)$$

effectively realizing the reverse diffusion via classical post-processing the forward diffusion data $\{\mathcal{O}^{(s)}\}_{s=1}^M$.

The statistical variance associated with $z_{0,i}$ scales as

$$\text{Var}[z_{0,i}] \sim \frac{\|P_i\|_{\text{sh}}^2}{M}, \quad (44)$$

where $\|P\|_{\text{sh}}^2$ denotes the *shadow norm* and is given by

$$\|P\|_{\text{sh}}^2 = \frac{\text{Tr}(P\mathcal{M}_t^{-1}(P))}{\text{Tr}(\mathcal{M}_t^{-1}(\mathbf{1}))} = \frac{w_{\mathcal{M}_t}(\mathbf{1})}{w_{\mathcal{M}_t}(P)}. \quad (45)$$

Therefore, to achieve a target variance ϵ^2 for a given Pauli observable P , the required sample size scales as

$$M \sim \frac{\|P\|_{\text{sh}}^2}{\epsilon^2}. \quad (46)$$

Physically, the Pauli weight $w_{\mathcal{M}_t}(P)$ quantifies the rate that measurement extracts information about the Pauli observable P . In reconstruction, the expectation value of P must be reweighted by $1/w_{\mathcal{M}_t}(P)$, which necessarily amplifies statistical fluctuations. A smaller Pauli weight indicates less information per sample and thus a larger amplification factor and a larger sample complexity.

The shadow norm exhibits characteristic time dependence: for short times ($t \rightarrow 0$), it diverges as $\mathcal{O}(1/t)$, reflecting that weak measurements initially acquire little information about $\bar{\rho}_0$, hence requiring many samples for accurate reconstruction. As t increases, the information per trajectory grows linearly before saturating. In the long-time limit ($t \rightarrow \infty$), the shadow norm converges to $3^{|P|}$, matching the scaling of projective Pauli measurements.

C. Petz Recovery

1. Global Petz Recovery

At the average state level, reversing quantum diffusion amounts to inverting the weak measurement channel $\mathcal{O}_t$ that governs the evolution of the average state $\bar{\rho}_t$. A general, near-optimal method for channel inversion is the *Petz recovery map* [28], which guarantees exact recovery when $\mathcal{F}_t$ is reversible on the support of $\bar{\rho}_0$ and provides strong theoretical performance guarantees for approximate cases.

For an infinitesimal time step dt , the Petz recovery map can be expressed in the “twirled” integral form

$$\begin{aligned} \tilde{\mathcal{R}}_{dt}(\sigma) &= \int_{-\infty}^{\infty} f(\tau) \mathcal{R}_{dt}^{\tau}(\sigma) d\tau, \\ \mathcal{R}_{dt}^{\tau}(\sigma) &= \bar{\rho}_t^{\frac{1-i\tau}{2}} \mathcal{F}_{dt}^{\dagger} \left(\bar{\rho}_{t+dt}^{\frac{-1+i\tau}{2}} \sigma \bar{\rho}_{t+dt}^{\frac{-1-i\tau}{2}} \right) \bar{\rho}_t^{\frac{1+i\tau}{2}}, \end{aligned} \quad (47)$$

where $f(\tau) = \frac{1}{2(\cosh(\pi\tau)+1)}$ and $\mathcal{F}_{dt}^{\dagger} = e^{dt\mathcal{L}^{\dagger}}$ denotes the adjoint channel of $\mathcal{F}_{dt}$, and the Liouvillian super-operator $\mathcal{L} = \mathcal{L}^{\dagger}$ is self-adjoint as defined in Eq. (14). Physically,

Eq. (47) describes how to undo the decoherence from $\mathcal{F}_{dt}$ using the eigenbasis of $\bar{\rho}_t$ as a prior. The forward channel $\mathcal{F}_{dt}$ slightly suppresses off-diagonal elements of $\bar{\rho}_t$ in its eigenbasis, corresponding to a drift toward the maximally mixed state. The Petz map applies the opposite drift by reweighting the eigencomponents of σ relative to $\bar{\rho}_t$, thereby restoring the lost coherence. Iterating $\tilde{\mathcal{R}}_{dt}$ over the full evolution time yields a reconstruction of the initial average state from the final diffused state.

2. Local Petz Recovery

While the *global* Petz map is an exact recovery channel in theory, it is generally intractable for large many-body systems due to its fully nonlocal action. A practical alternative arises when the ensemble exhibits predominantly short-range correlations, characterized by a finite *Markov length* ξ : correlations between a region A and the distant complement C decay rapidly once a buffer region B (of width ξ) is included. Formally, this corresponds to the conditional mutual information (CMI) $I_{\rho}(A : C|B)$ being small for appropriate tripartitions (A, B, C) . In this near-Markovian regime, the global state is well-approximated by a quantum Markov chain, and one can replace the global Petz map by a *local* version acting only on $S_j = A \cup B$.

The local Petz map $\tilde{\mathcal{R}}_{dt, S_j}$ is simply constructed by restricting the twirled Petz formula Eq. (47) to S_j , using the reduced density matrix ρ_{t, S_j} as the prior and $\mathcal{F}_{dt, S_j}$ as the local measurement channel. Overlapping regions $\{S_j\}$ are chosen to cover the system, and the local recoveries $\tilde{\mathcal{R}}_{dt, S_j}$ are applied in sequence or in parallel following the scheme of Ref. 29. This yields a finite-depth recovery circuit implementable on quantum hardware, with accuracy controlled by the residual CMI.

Theorem 2 (Local Petz Recovery with Finite Markov Length). *Let $\bar{\rho}_0$ be an n -qubit state evolved under the infinitesimal measurement channel $\mathcal{F}_{dt}$ for total time $T = \mathcal{O}(n)/\gamma$, producing $\bar{\rho}_T = \mathcal{F}_T(\bar{\rho}_0)$. Suppose that for every intermediate $\bar{\rho}_t$ and every subsystem $S_j = A \cup B$, the CMI satisfies $I_{\bar{\rho}_t}(A : C|B) \leq \delta$, where C is the complement of S_j . Then, applying N_T sequential local Petz maps $\tilde{\mathcal{R}}_{dt, S_j}$ over a covering $\{S_j\}$ produces a final state $\bar{\rho}'_0$ satisfying*

$$\|\bar{\rho}'_0 - \bar{\rho}_0\|_1^2 \leq \frac{2 \ln 2 \cdot \mathcal{O}(n)}{\gamma} \cdot \delta + E_{\text{diff}}(T), \quad (48)$$

with $E_{\text{diff}}(T)$ decaying exponentially as the diffusion time T increases.

Proof Sketch. The result follows from [29] by bounding the single-step recovery error via the CMI, then summing over N_T steps to obtain a telescoping bound. The data-processing inequality ensures CMI monotonicity, so the total error is controlled by δ . Meanwhile, the residual term $E_{\text{diff}}(T)$ decays exponentially with T as repeated

weak measurements diffuse the state towards a maximally mixed product state by rapidly damping higher-weight Pauli operators as in Eq. (21).

If the forward weak measurement schedule is arranged so that qubits measured in consecutive steps are separated by a distance much greater than the Markov length ξ , the forward channel $\mathcal{F}_t$ can be implemented as a constant-depth circuit. The local Petz maps, restricted to their respective S_j , inherit the same finite-depth structure, enabling efficient implementation.

The following protocol constructs local Petz recovery maps from forward-diffusion measurement data and implements them as local quantum channels on quantum hardware:

- (i) *Collect data.* Repeatedly prepare the unknown target average state $\bar{\rho}_0$ and apply the forward diffusion process via randomized weak measurements. Record the complete measurement record $\mathcal{O} = \{(O_t, o_t)\}$ for each trajectory, forming a dataset $\{\mathcal{O}\}$.
- (ii) *Process data.* Apply weak-measurement classical shadow tomography to the dataset $\{\mathcal{O}\}$ to reconstruct an estimate $\tilde{\rho}_0$ of the initial average state. Using this estimate, infer the intermediate states $\tilde{\rho}_t = \mathcal{F}_t(\tilde{\rho}_0)$ via classical simulation of the known forward channel $\mathcal{F}_t$.
- (iii) *Compute local Petz maps.* For each overlapping local region S_j (with size exceeding the Markov length ξ), construct the local Petz recovery map $\tilde{\mathcal{R}}_{dt, S_j}$ by restricting the *twirled Petz* formula Eq. (47) to S_j , using the reduced state $\tilde{\rho}_{t, S_j}$ and the corresponding local channel $\mathcal{F}_{dt, S_j}$. Arrange and stack these local maps in the reverse order of the forward schedule to obtain the overall Petz recovery protocol $\tilde{\mathcal{R}}_t$ (see [29] for the detailed construction).
- (iv) *Implement reverse diffusion.* Each step of $\tilde{\mathcal{R}}_t$ is a local quantum channel that can, in principle, be implemented on quantum hardware. Initialize the device in a state $\bar{\rho}'_T$ (e.g., a random product state evolved forward to time T), then apply $\bar{\rho}'_0 = \tilde{\mathcal{R}}_T(\bar{\rho}'_T)$ to obtain a recovered approximation to the original state $\bar{\rho}_0$. This procedure enables repeated state preparation directly on the quantum device.

This protocol provides a systematic construction of the reverse process as a composition of local quantum channels, tailored for forward diffusion driven by randomized measurements. In contrast to prior approaches to average-state recovery based on learned reverse quantum channel [4, 6–8, 11, 13, 14], the Petz recovery approach requires no learning: the recovery maps are constructed directly from data and knowledge of the forward channel. This observation offers an important theoretical guarantee for learning-based methods—namely, that

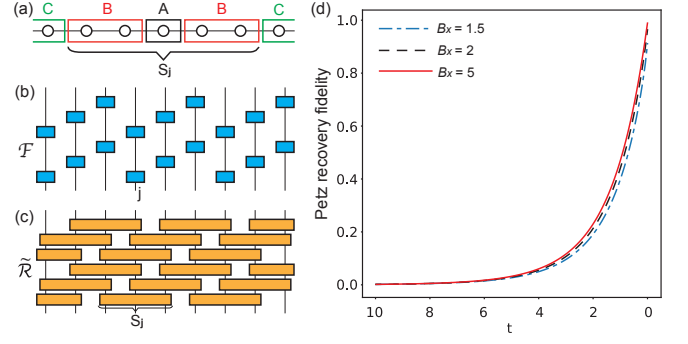


FIG. 5. (a) Regions A (qubit j), B , C and S_j . When adjacent steps measure far-apart qubits, the circuit of (b) the forward weak measurement channel $\mathcal{F}$ and (c) the Petz recovery channel $\tilde{\mathcal{R}}$ can be rearranged into a finite depth circuit. (d) Fidelity of Petz recovery at time t with the initial state.

a purely classical algorithm exists to build local recovery maps from measurement data. Therefore, any successful learning-based recovery can be viewed as approximating, in principle, a well-defined recovery process given by local Petz maps.

3. Numerical Demonstration

To illustrate the proposed protocol, we perform a numerical experiment on a 10-qubit transverse-field Ising chain with open boundary conditions,

$$H = -J \sum_i \sigma_{z,i} \sigma_{z,i+1} - B_x \sum_i \sigma_{x,i}, \quad (49)$$

using $J = 1.0$, and $B_x = 1.5, 2.0$ and 5.0 as in Fig. 5. This choice places the system well away from the critical point ($B_x = 1.0$), ensuring a finite correlation length and a ground state close to the product state $|+\rangle^{\otimes n}$. The chain is initialized in the ground state $\bar{\rho}_0$ of H and evolved under randomized weak-measurement dynamics for a total time $T = 10$ with step size $dt = 0.01$ (1000 diffusion steps).

Local Petz maps $\tilde{\mathcal{R}}_{dt, S_j}$ are constructed for all contiguous three-qubit regions S_j (each exceeding the Markov length) and arranged in a constant-depth circuit, ensuring that the entire protocol inverts the forward diffusion approximately while remaining finite in depth. The recovered state $\bar{\rho}'_0$ achieves fidelities of 0.911, 0.963, and 0.989 with respect to $\bar{\rho}_0$ for $B_x = 1.5, 2.0$, and 5.0 , respectively, as in Fig. 5.

4. Petz Recovery as Time-Reversed Diffusion

We establish that the Petz recovery map provides a quantum generalization of classical time-reversed diffusion under weak measurements, and that in the classical large-spin limit the two become equivalent.

Consider an n -qubit separable state (no quantum entanglement), expressed as a mixture of tensor product of spin coherent states

$$\bar{\rho}_t = \int d\mathbf{n} p(\mathbf{n}, t) |\mathbf{n}\rangle\langle\mathbf{n}|, \quad (50)$$

where the spin configuration is described by $\mathbf{n} = (\mathbf{n}_1, \dots, \mathbf{n}_n)$ with $|\mathbf{n}_j| = 1$, and the corresponding coherent state reads

$$|\mathbf{n}\rangle\langle\mathbf{n}| = \prod_{j=1}^n \frac{\mathbb{1} + \mathbf{n}_j \cdot \boldsymbol{\sigma}_j}{2}, \quad (51)$$

and $p(\mathbf{n}, t)$ is a normalized probability density over the n -fold product of Bloch spheres.

Consider randomized Pauli weak measurements, whose action is equivalent to applying independent depolarizing channels on each qubit. Since such local measurements do not generate entanglement, every trajectory remains a pure product state throughout the process. Consequently, the ensemble evolution is fully captured by the probability density $p(\mathbf{n}, t)$ over coherent state configurations $\mathbf{n}$. The resulting stochastic dynamics yield the forward Fokker–Planck equation

$$\partial_t p(\mathbf{n}, t) = \frac{2\gamma}{3n} \nabla_{\perp}^2 p(\mathbf{n}, t), \quad (52)$$

where $\nabla_{\perp}^2 = \sum_{j=1}^n \nabla_{\perp, j}^2$ is the sum of Laplacians on individual Bloch spheres.

Given the forward diffusion history of the average state $\bar{\rho}_t = \mathcal{F}_t(\bar{\rho}_0)$, the reverse process can be implemented as

$$\bar{\rho}_0 = \mathcal{R}_t^0(\bar{\rho}_t), \quad (53)$$

where $\mathcal{R}_t^0$ is the (untwirled) Petz map derived from Eq. (47),

$$\mathcal{R}_t^0(\sigma) = \bar{\rho}_t^{-1/2} \mathcal{F}_t^\dagger(\bar{\rho}_{t+\text{dt}}^{-1/2} \sigma \bar{\rho}_{t+\text{dt}}^{-1/2}) \bar{\rho}_t^{-1/2}, \quad (54)$$

which reweights σ by $\bar{\rho}_t^{-1/2}$ through the adjoint channel. For $\sigma = \int q(\mathbf{n}, t) |\mathbf{n}\rangle\langle\mathbf{n}| d\mathbf{n}$, $\mathcal{R}_t^0(\sigma)$ induces a backward diffusion of the distribution $q(\mathbf{n}, t)$ — a quantum generalization of classical reverse diffusion. The PDE governing this backward evolution is given in SM [26].

In the large-spin (classical) limit, coherent states with different $\mathbf{n}$ become orthogonal, $\langle\mathbf{n}|\mathbf{n}'\rangle \rightarrow \delta_{\mathbf{n}, \mathbf{n}'}$. The Petz equation becomes the classical backward Fokker–Planck equation

$$\partial_t q(\mathbf{n}, t) = \frac{2\gamma}{3n} \nabla_{\perp}^2 q(\mathbf{n}, t) - \frac{4\gamma}{3n} \nabla_{\perp} \cdot [q(\mathbf{n}, t) \nabla_{\perp} \log p(\mathbf{n}, t)], \quad (55)$$

exactly reversing Eq. (52), with $\frac{4\gamma}{3n} \nabla_{\perp} \log p$ as the backward drift.

This result reveals a precise theoretical connection: in the classical limit, the Petz recovery map is time-reversed diffusion. This not only clarifies the physical meaning of Petz recovery, but also provides a rigorous bridge between quantum recovery channels and classical stochastic reversals, ensuring that any successful quantum Petz reconstruction has a well-defined classical counterpart.

IV. CONCLUSION AND DISCUSSION

We developed a measurement-based framework for quantum diffusion that unifies trajectory-wise and ensemble-average reversal. On the forward side, randomized weak measurements generate stochastic pure-state trajectories while inducing a measurement channel that is locally scrambled and diagonal in the Pauli basis. On the reverse side, we presented two complementary routes: (i) trajectory-level reversal via score-matching/control–Hamiltonian learning; and (ii) ensemble-level reversal via either Petz recovery or classical shadow reconstruction.

a. Summary of contributions. (i) We formalized trajectory-level reverse diffusion as a control problem and showed its equivalence to quantum score matching, yielding a principled objective for learning the reverse flow. (ii) For ensemble averages, we introduced a practical *local* Petz recovery when correlations are short-ranged, leveraging recoverability guarantees for Lindbladian dynamics and near-optimal Petz maps; this connects quantum channel inversion to reversing measurement-induced decoherence. (iii) We established a complementary *classical* post-processing route via classical shadow tomography, exploiting that our Pauli-twirled measurement-and-prepare channel is diagonal in the Pauli basis; its inverse gives an unbiased estimator of $\bar{\rho}_0$ with rigorous concentration bounds for local observables. (iv) Conceptually, we clarified that Petz recovery implements a structured quantum analogue of time-reversed diffusion and reduces to classical backward diffusion in the large-spin/separable limit, linking recovery channels to Fokker–Planck reversals. This complements recent diffusion-style generative models for quantum data that operate in either state space or density-matrix space.

b. Broader implications. The measurement-driven formulation naturally couples data acquisition and control: the same weak-measurement records that drive forward diffusion also provide conditioning signals for reverse synthesis. Local Petz maps offer a hardware-friendly, finite-depth realization of ensemble reversal under finite Markov length, whereas classical shadows provide an efficient, device-agnostic alternative when the goal is inference rather than state preparation. Together, these tools place quantum diffusion models on firmer mathematical footing and connect them to established theories of recoverability and randomized tomography.

c. Limitations and outlook. Our trajectory-wise reversal presently relies on classical decoders (simulation- or learning-based) to infer sufficient state information for feedback, a key difference from classical denoising where the state is directly observable. Scaling such decoders, tightening end-to-end error bounds (from Hamiltonian learning error to ensemble Wasserstein/trace-distance guarantees), and extending beyond randomized Pauli measurements to other locally scrambled schemes (e.g., Clifford- or unitary-2-design-based protocols) are immediate next steps. It is also promising to explore recovery-

aware measurement schedules and noise models that optimize sample complexity and circuit depth, and to interface these methods with structure-preserving diffusion for mixed states and with recent progress on designing open dynamics with prescribed steady states.

Overall, this work provides a concise blueprint for quantum diffusion: use weak measurements to generate and learn from forward randomness; reverse trajectories via learned control when needed; and reverse ensembles via Petz recovery or classical shadows when sufficient statistics are available. We expect these ingredients to be directly useful in near-term platforms that already support weak, randomized measurements and local quantum channels.

Note Added: Upon the completion of our work, we became aware of a complementary work by Hu et al. [30], which also discussed local recovery mechanisms in diffusion processes. Their work also established a quantum-classical correspondence between Petz recovery maps and classical diffusion models, and uses conditional mutual information to determine the possibility of local denoising.

While our work focuses on quantum state generation, their work addresses classical data generation through local denoising.

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Supplemental Material

Appendix A: Derivations for the Score Matching of the General Diffusion Dynamics

This appendix provides a detailed, self-contained derivation of the reverse-time dynamics for a general diffusion process and its application to the stochastic evolution of quantum states under weak measurement. We begin by outlining the general theory of stochastic differential equations (SDEs) and score-based modeling. We then connect this mathematical framework to the physical process of weak measurement. Finally, we establish a formal equivalence between learning the reverse diffusion process for an ensemble of pure states and learning an effective time-reversed unitary evolution.

1. General Diffusion Dynamics

a. The generic SDE, Fokker–Planck PDE, and ODE Descriptions

A generalized diffusion process for a state vector $\mathbf{z}(t)$ in an N dimensional space can be described by a stochastic differential equation (SDE) with a M -dimensional and state-dependent noise. The generic SDE takes the form (in which all quantities are real valued):

$$d\mathbf{z}_t = \mathbf{f}(\mathbf{z}_t, t)dt + \mathbf{G}(\mathbf{z}_t, t)d\mathbf{w}_t, \quad (\text{A1})$$

where:

- $\mathbf{z}_t$ is an N -dimensional **state vector** representing the system's state at time t .
- $\mathbf{f}(\mathbf{z}_t, t)$ is an N -dimensional **drift field** (drift force), describing the deterministic part of the evolution.
- $\mathbf{G}(\mathbf{z}_t, t)$ is an $N \times M$ matrix representing the **noise amplitude**, which couples the noise sources to the state vector.
- $d\mathbf{w}_t = (dw_{t,1}, \dots, dw_{t,M})^T$ is an M -dimensional vector representing a Wiener process **noise**, satisfying the a probability distribution (usually Gaussian) with the first and second moments ($\mathbb{E}$ stands for mean value):

$$\mathbb{E}[dw_{t,i}] = 0, \quad \mathbb{E}[dw_{t,i}dw_{t,j}] = \delta_{ij}dt.$$

The time evolution of the probability distribution $p(\mathbf{z}, t)$ of the state vector $\mathbf{z}$ is known to be governed by the corresponding Fokker–Planck partial differential equation (PDE):

$$\partial_t p(\mathbf{z}, t) = -\nabla \cdot [\mathbf{f}(\mathbf{z}, t)p(\mathbf{z}, t)] + \frac{1}{2} \partial_i \partial_j (\Sigma_{ij}(\mathbf{z}, t)p(\mathbf{z}, t)), \quad (\text{A2})$$

where Einstein summation convention over repeated indices are assumed, and we have introduced the $N \times N$ noise covariance matrix

$$\Sigma(\mathbf{z}, t) = \mathbf{G}(\mathbf{z}, t)\mathbf{G}^T(\mathbf{z}, t). \quad (\text{A3})$$

Using the probability distribution $p(\mathbf{z}, t)$, the SDE of state vector $\mathbf{z}_t$ is statistically equivalent to a deterministic ordinary differential equation (ODE), known as the probability flow ODE:

$$d\mathbf{z}_t = \mathbf{n}(\mathbf{z}_t, t)dt, \quad (\text{A4})$$

with the velocity field $\mathbf{n} = (v_1, \dots, v_N)^T$ given by:

$$v_i(\mathbf{z}, t) = f_i(\mathbf{z}, t) - \frac{1}{2} \left[\partial_j \Sigma_{ij}(\mathbf{z}, t) + \Sigma_{ij}(\mathbf{z}, t) \partial_j \log p(\mathbf{z}, t) \right], \quad (\text{A5})$$

or more concisely in vector form,

$$\mathbf{n}(\mathbf{z}, t) = \mathbf{f}(\mathbf{z}, t) - \frac{1}{2} \left[\nabla \cdot \Sigma(\mathbf{z}, t) + \Sigma(\mathbf{z}, t) \cdot \nabla \log p(\mathbf{z}, t) \right], \quad (\text{A6})$$

where $(\nabla \cdot \Sigma)_i = \partial_j \Sigma_{ij}$.

b. Reverse Diffusion and Score Matching

A key result from stochastic calculus is that the forward diffusion process described by Eq. (A1) has a corresponding reverse-time process. The reverse SDE, which allows for generating samples from the distribution $p(\mathbf{z}, 0)$ by starting from a simple prior at time T , at time $\tilde{t} = T - t$ is given by:

$$d\mathbf{z}_{\tilde{t}} = [-\mathbf{f}(\mathbf{z}_{\tilde{t}}, \tilde{t}) + \nabla \cdot \mathbf{\Sigma}(\mathbf{z}_{\tilde{t}}, \tilde{t}) + \mathbf{\Sigma}(\mathbf{z}_{\tilde{t}}, \tilde{t}) \cdot \nabla \log p(\mathbf{z}_{\tilde{t}}, \tilde{t})] d\tilde{t} + \mathbf{G}(\mathbf{z}_{\tilde{t}}) d\mathbf{w}_{\tilde{t}} . \quad (\text{A7})$$

where $d\mathbf{w}_{\tilde{t}}$ is a standard Wiener process in the reverse time direction.

The generative power of this framework hinges on knowledge of the score function incorporating the information of $\nabla \log p(\mathbf{z}, t)$. In score-based modeling, we define a learnable model $s_{\theta}(\mathbf{z}, t)$ with variational parameters θ to approximate the score-dependent part of the reverse drift:

$$s_{\theta}(\mathbf{z}, t) \approx \nabla \cdot \mathbf{\Sigma}(\mathbf{z}, t) + \mathbf{\Sigma}(\mathbf{z}, t) \cdot \nabla \log p(\mathbf{z}, t) , \quad (\text{A8})$$

by which one can implement the reverse SDE in Eq. (A7) by the score function $s_{\theta}(\mathbf{z}, t)$. The model is trained by minimizing the score-matching loss, which is the expected squared deviation between the model and the true score term:

$$\mathcal{L}_{\theta} = \mathbb{E}_{\mathbf{z}_t \sim p(\mathbf{z}, t)} \left[\|s_{\theta}(\mathbf{z}_t, t) - (\nabla \cdot \mathbf{\Sigma})(\mathbf{z}_t) - \mathbf{\Sigma}(\mathbf{z}_t) \nabla \log p(\mathbf{z}, t)\|^2 \right] \quad (\text{A9})$$

Directly computing this loss is often intractable. A practical and equivalent alternative is denoising score matching. Given a forward step $\mathbf{z}_{t+dt} = \mathbf{z}_t + \mathbf{f}(\mathbf{z}_t)dt + \mathbf{g}(\mathbf{z}_t)d\mathbf{w}_t$, it can be shown that the above loss function in Eq. (A9) is equivalent to the following denoising loss function up to a constant:

$$\mathcal{L}_{\theta} = \frac{1}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left[\left\| s_{\theta}(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t)dt, t) + \frac{\mathbf{z}_{t+dt} - \mathbf{z}_t - \mathbf{f}dt}{dt} \right\|^2 \right] . \quad (\text{A10})$$

This objective trains the model to predict the noise term that corrupted the state, thereby implicitly learning the score without needing to compute it directly.

Lastly, starting from a backward probability distribution $q(\mathbf{z}, \tilde{t})$, the backward diffusion PDE driven by the forward probability distribution $p(\mathbf{z}, \tilde{t})$ takes the form

$$\partial_{\tilde{t}} q(\mathbf{z}, \tilde{t}) = -\nabla \cdot [(-\mathbf{f}(\mathbf{z}, \tilde{t}) + \nabla \cdot \mathbf{\Sigma}(\mathbf{z}, \tilde{t}) + \mathbf{\Sigma}(\mathbf{z}, \tilde{t}) \cdot \nabla \log p(\mathbf{z}, \tilde{t})) q(\mathbf{z}, \tilde{t})] + \frac{1}{2} \partial_i \partial_j (\Sigma_{ij}(\mathbf{z}, \tilde{t}) q(\mathbf{z}, \tilde{t})) , \quad (\text{A11})$$

where $\nabla \cdot \mathbf{\Sigma}(\mathbf{z}, \tilde{t}) + \mathbf{\Sigma}(\mathbf{z}, \tilde{t}) \cdot \nabla \log p(\mathbf{z}, \tilde{t})$ plays the role of a drifting force towards the distribution $p(\mathbf{z}, \tilde{t})$.

2. The nonlinear SDE from the Weak Measurement Protocol

a. The Weak Measurement Protocol

We consider a quantum system initially prepared in the state ρ_0 . Ancilla-assisted weak measurement of an observable O_t is implemented by coupling the system to an ancilla qubit, prepared in $|0\rangle_A$, via the interaction Hamiltonian

$$H = \lambda O_t \sigma_A^y . \quad (\text{A12})$$

After evolving for a finite time δt , the ancilla is projectively measured in the σ_A^x basis, yielding an outcome o_t . In the continuous limit, where $\delta t \rightarrow 0$ and $\lambda \rightarrow \infty$ such that the measurement rate $\gamma = \lambda^2 \delta t$ is constant, this interaction is described by the Kraus operator:

$$K_{\delta t}(O_t, \delta o_t) = \frac{1}{\sqrt{N_{O_t}}} \exp\left(-\gamma \delta t + 2\gamma O_t \delta o_t\right), \quad (\text{A13})$$

where N_{O_t} is a normalization factor such that $\sum_{\delta o_t} K_{\delta t}^{\dagger}(O_t, \delta o_t) K_{\delta t}(O_t, \delta o_t) = \mathbb{1}$.

b. Forward Measurement-Based Diffusion

Assuming the observable $O_t = O_t^\dagger$ is Hermitian, the corresponding weak measurement of observable O_t in time δt with measurement outcome o_t will update the system's state via a Kraus operator $K_{\delta t}(O_t, \delta o_t)$ into:

$$\rho_{t+dt} = \frac{K_{\delta t}(O_t, \delta o_t) \rho_t K_{\delta t}^\dagger(O_t, \delta o_t)}{\text{Tr} \left[K_{\delta t}(O_t, \delta o_t) \rho_t K_{\delta t}^\dagger(O_t, \delta o_t) \right]} . \quad (\text{A14})$$

The probability of observing outcome o_t is state-dependent, and the mean value of o_t from weak measurement defined above is given by

$$\mathbb{E}[o_t] = 2\sqrt{\gamma\delta t} \langle O_t \rangle , \quad (\text{A15})$$

where we denote $\langle O_t \rangle = \langle \psi_t | O_t | \psi_t \rangle$ for short. For instance, for one step of weak measurement of spin 1/2 operator O_t with outcomes $o_t = \pm 1$ in time dt , the probability of observing outcome o_t is:

$$p(o_t | \rho_t) = \text{Tr} \left[K_{\delta t}(O_t, \delta o_t) \rho_t K_{\delta t}^\dagger(O_t, \delta o_t) \right] = \frac{1}{2} + o_t \sqrt{\gamma\delta t} \langle O_t \rangle = \frac{1}{2} + 2\gamma \langle O_t \rangle \delta o_t , \quad \delta o_t = \frac{1}{2} \sqrt{\frac{\delta t}{\gamma}} o_t . \quad (\text{A16})$$

c. The Stochastic Process from Weak Measurement

The evolution of the quantum state ρ_t is found by expanding the update rule in Eq. (A14) to the first order in δt (for the derivations that follow, we denote the time increment as $\delta t = dt$ for convenience), after which we arrive at the nonlinear SDE for the normalized state:

$$d\rho_t = -\frac{\gamma}{2} [O_t, [O_t, \rho_t]] dt + \sqrt{\gamma} \{O_t - \langle O_t \rangle, \rho_t\} dw_t \quad (\text{A17})$$

where

$$dw_t = o_t \sqrt{dt} - 2\sqrt{\gamma} dt \langle O_t \rangle = 2\sqrt{\gamma} (do_t - \langle O_t \rangle dt) \quad (\text{A18})$$

can be treated as a standard Wiener noise term satisfying $\mathbb{E}[dw_t] = 0$ and $\mathbb{E}[dw_t^2] = dt$. The first term in Eq. (A17) is a deterministic drift representing decoherence, while the second is a stochastic term describing the random walk from quantum state collapse. If the state is pure, $\rho_t = |\psi_t\rangle\langle\psi_t|$, this process preserves purity, and the nonlinear SDE in Eq. (A17) is equivalent to the following evolution of the quantum state:

$$d|\psi_t\rangle = \left(-\frac{\gamma}{2} (O - \langle O \rangle)^2 dt + \sqrt{\gamma} (O - \langle O \rangle) dw_t \right) |\psi_t\rangle . \quad (\text{A19})$$

d. The nonlinear SDE in the Pauli basis

The nonlinear SDE for the weak measurement in Eq. (A17) can be written more explicitly by expanding the n -qubit density matrix in the Pauli basis:

$$\rho_t = \frac{1}{2^n} \sum_i z_i P_i = \frac{1}{2^n} \mathbf{z} \cdot \mathbf{P} , \quad (\text{A20})$$

where P_i ($1 \leq i \leq N$) runs over all the $N = 4^n$ Pauli string operators, with $P_0 = I$ being the identity operator, $\mathbf{P} = (P_1, \dots, P_N)^T$, and $z_i = \text{Tr}(\rho P_i)$. This defines a state vector $\mathbf{z} = (z_1, \dots, z_N)^T = \text{Tr}(\rho \mathbf{P})$ for the density matrix ρ . By taking trace of the nonlinear SDE in Eq. (A17) with vector operator $\mathbf{P}$, and denoting the state vector at time t by $\mathbf{z}_t$, we can rewrite it into a nonlinear SDE in the Pauli basis as:

$$d\mathbf{z}_t = \mathbf{f}(\mathbf{z}_t, t) dt + \mathbf{g}(\mathbf{z}_t, t) dw_t , \quad (\text{A21})$$

where the drift vector $\mathbf{f}(\mathbf{z}, t)$ and the noise vector $\mathbf{g}(\mathbf{z}, t)$ are defined by

$$\mathbf{f}(\mathbf{z}_t, t) = -\frac{\gamma}{2} \text{Tr}([O_t, [O_t, \rho_t]] \mathbf{P}) , \quad \mathbf{g}(\mathbf{z}_t, t) = \sqrt{\gamma} \text{Tr}(\{O_t - \langle O_t \rangle, \rho_t\} \mathbf{P}) , \quad (\text{A22})$$

Let $\mathbf{P} = \{P_i\}_{i=1}^{4^n}$ denote the n -qubit Pauli-operator basis with $\text{Tr}(P_i P_j) = 2^n \delta_{ij}$. Any state ρ_t or observable O_t can be expanded as

$$\rho_t = \frac{\mathbf{z}_t \cdot \mathbf{P}}{2^n}, \quad O_t = \mathbf{x}_t \cdot \mathbf{P}, \quad (\text{A23})$$

with $z_{t,i} = \text{Tr}(\rho_t P_i) = \langle \psi_t | P_i | \psi_t \rangle$, $x_{t,i} = \text{Tr}(O_t P_i)/2^n$. Then Eq. (A22) becomes (Einstein summation convention implied)

$$\begin{aligned} f_l(\mathbf{z}_t, t) &= -\gamma(c_{ijm}c_{klm} - c_{ikm}c_{jlm})x_{t,i}x_{t,j}z_{t,k}, \\ g_l(\mathbf{z}_t, t) &= \sqrt{\gamma}((c_{ijl} + c_{jil})x_{t,i}z_{t,j} - 2x_{t,i}z_{t,i}z_{t,l}), \end{aligned} \quad (\text{A24})$$

and $c_{ijk} = \text{Tr}(P_i P_j P_k)/\text{Tr} \mathbf{1}$ being the operator product expansion (OPE) coefficient.

Compared with the generic SDE in Eq. (A1), our weak measurement nonlinear SDE has $M = 1$ dimensional noise dw_t (which is a number) introduced by weak measurement in the $\mathbf{x}_t$ direction, and the matrix $\mathbf{G}(\mathbf{z}, t)$ reduces to the noise vector $\mathbf{g}(\mathbf{z}, t)$ here. Accordingly, we define the probability distribution of the state ρ in terms of state vector $\mathbf{z}$ as $p(\mathbf{z}, t)$.

All the forward/backward SDE, ODE, PDE, score function and loss function specifically for our weak measurement model then follows from App. A1 by setting the noise dimension $M = 1$.

3. The Unitary Reverse Procedure for Pure States

a. From learning unitary to learning score function

For an ensemble of pure states, the forward diffusion process implemented by weak measurement must be purity-preserving. As is clear from Eq. (A19), a pure state remains pure along the quantum trajectory. Consequently, the time-evolution process is a unitary evolution.

For such pure state evolutions, we are therefore motivated to define a unitary score function being an infinitesimal time- dt unitary evolution $V_\theta(\mathbf{z}, t)$ (for some fixed small time step dt), such that it approximates the infinitesimal forward evolution

$$\rho_{t+dt} = \frac{K_{dt}(O_t, do_t) \rho_t K_{dt}^\dagger(O_t, do_t)}{\text{Tr} [K_{dt}(O_t, do_t) \rho_t K_{dt}^\dagger(O_t, do_t)]} \approx V_\theta(\mathbf{z}_t, t) \rho_t V_\theta^\dagger(\mathbf{z}_t, t) = |\psi_{t+dt}\rangle \langle \psi_{t+dt}|, \quad \rho_t = |\psi_t\rangle \langle \psi_t|. \quad (\text{A25})$$

Once this score function is learned, the time-reversed process can be generated by the inverse unitary transformations $V_\theta^\dagger(\mathbf{z}_t, t)$, which gives a purity-preserving reverse unitary evolution.

We are thus straightforwardly motivated to define a loss function

$$\begin{aligned} \mathcal{L}_\theta &= \frac{1}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left\| V_\theta(\mathbf{z}_{t+dt}, t) \rho_t V_\theta^\dagger(\mathbf{z}_{t+dt}, t) - \frac{K_{dt}(O_t, do_t) \rho_t K_{dt}^\dagger(O_t, do_t)}{\text{Tr} [K_{dt}(O_t, do_t) \rho_t K_{dt}^\dagger(O_t, do_t)]} \right\|_F^2 \\ &= \frac{1}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left\| V_\theta(\mathbf{z}_{t+dt}, t) \rho_t V_\theta^\dagger(\mathbf{z}_{t+dt}, t) - \rho_{t+dt} \right\|_F^2 \\ &= 1 - \mathbb{E}_{\psi_t, \psi_{t+dt}} |\langle \psi_{t+dt} | V_\theta(\mathbf{z}_{t+dt}, t) | \psi_t \rangle|^2, \end{aligned} \quad (\text{A26})$$

optimizing which gives the desired unitary score V_θ . We now show that this is equivalent to the denoising loss function in Eq. (A10) usually defined in the diffusion problem. Using the fact that

$$\|\mathbf{z} \cdot \mathbf{P}\|^2 = \sum_{ij} z_i z_j \text{tr}(P_i P_j) = 2^n \sum_i z_i^2 = 2^n \|\mathbf{z}\|^2, \quad (\text{A27})$$

the denoising loss function in Eq. (A10) can be rewritten as

$$\begin{aligned} \mathcal{L}_\theta &= \frac{2^n}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left\| [s_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t)dt, t) - \mathbf{f}(\mathbf{z}_t)] \cdot \frac{\mathbf{P}}{2^n} + \left(\frac{\mathbf{z}_{t+dt} - \mathbf{z}_t}{dt} \right) \cdot \frac{\mathbf{P}}{2^n} \right\|^2 \\ &= \frac{2^n}{2} \mathbb{E}_{\mathbf{z}_t, \mathbf{z}_{t+dt}} \left\| [s_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t)dt, t) - \mathbf{f}(\mathbf{z}_t)] \cdot \frac{\mathbf{P}}{2^n} + \frac{\rho_{t+dt} - \rho_t}{dt} \right\|^2. \end{aligned} \quad (\text{A28})$$

Comparing with Eq. (A26), we find if we make the following identification:

$$[\mathbf{s}_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t)dt, t) - \mathbf{f}(\mathbf{z}_t)] \cdot \frac{\mathbf{P}}{2^n} = \frac{\rho_t - V_\theta(\mathbf{z}_{t+dt}, t) \rho_t V_\theta^\dagger(\mathbf{z}_{t+dt}, t)}{dt}, \quad (\text{A29})$$

then the loss function $\mathcal{L}_\theta$ we defined in Eq. (A26) is up to a global factor equivalent to the denoising loss function in Eq. (A10)!

Further, to the linear order of dt , the infinitesimal unitary V_θ can be parametrized by

$$V_\theta = e^{iH_\theta dt} \approx I + iH_\theta dt, \quad H_\theta = \sum_j \eta_{\theta,j} P_j = \boldsymbol{\eta}_\theta \cdot \mathbf{P}, \quad (\text{A30})$$

where H_θ is a Hermitian generator. (Keeping only up to the first order in dt of V_θ because our SDE is accurate in the first order in dt .) Thus, $V_\theta \rho_t V_\theta^\dagger - \rho_t \approx i[H_\theta, \rho_t]dt$, and we can identify the following:

$$\begin{aligned} [\mathbf{s}_\theta(\mathbf{z}_{t+dt} - \mathbf{f}(\mathbf{z}_t)dt, t) - \mathbf{f}(\mathbf{z}_t)] \cdot \frac{\mathbf{P}}{2^n} &= -i[H_\theta, \rho_t] = -\frac{i}{2^n} [\boldsymbol{\eta}_\theta \cdot \mathbf{P}, \mathbf{z}_t \cdot \mathbf{P}] \\ &= -\frac{i}{2^n} \sum_{jk} [P_j, P_k] \eta_{\theta,j} z_{t,k}. \end{aligned} \quad (\text{A31})$$

We see that $V_\theta = e^{iH_\theta}$ encodes the information of the loss function $\mathbf{s}_\theta - \mathbf{f}$.

b. Error bound for Unitary Learning

The equivalence between learning the score function and learning a unitary generator allows us to bound the error of the generative model. Here we consider the error bound of the unitary reverse generative model.

First, for a pair of pure states $|\phi\rangle$ and $|\psi\rangle$, we define their trace distance as

$$D(|\phi\rangle, |\psi\rangle) = \text{Tr} \sqrt{(\rho_\phi - \rho_\psi)^2} = \sqrt{2(1 - |\langle\phi|\psi\rangle|^2)}, \quad \rho_\phi = |\phi\rangle\langle\phi|, \quad \rho_\psi = |\psi\rangle\langle\psi|. \quad (\text{A32})$$

Assume the learned probability distribution from the unitary reverse generative model is p_θ , and the true probability distribution of pure states p_{true} . We then define the **p-Wasserstein distance** with trace distance cost is defined by [8]

$$W_p(p_\theta, p_{\text{true}}) = \left(\inf_{\pi \in \Pi(p_\theta, p_{\text{true}})} \int D(|\phi\rangle, |\psi\rangle)^p \pi(|\phi\rangle, |\psi\rangle) d\phi d\psi \right)^{\frac{1}{p}}, \quad (\text{A33})$$

where $\Pi(p_\theta, p_{\text{true}})$ runs over all the **couplings** (joint distributions) π on pairs of states $(|\phi\rangle, |\psi\rangle)$ whose marginals are p_θ and p_{true} , namely satisfying $\int \pi(|\phi\rangle, |\psi\rangle) d\phi = p_{\text{true}}(|\psi\rangle)$, and $\int \pi(|\phi\rangle, |\psi\rangle) d\psi = p_\theta(|\phi\rangle)$.

We now show the following theorem:

Theorem 3. Distribution Convergence Theorem. *Let p_{true} be the true probability distribution of initial pure states, and p_θ denote the learned probability distribution obtained after a reverse diffusion process of total evolution time $T = \mathcal{O}(n)/\gamma$ generated by the learned control unitary V_θ . Assume the learned control unitary satisfies*

$$\mathbb{E}_{\psi_t, \psi_{t+dt}} \sqrt{1 - |\langle\psi_{t+dt}|V_\theta(\mathbf{z}_{t+dt}, t)|\psi_t\rangle|^2} \leq \epsilon dt, \quad (\text{A34})$$

where $\mathbf{z}_t$ is the state vector of $|\psi_t\rangle$. Also, assume V_θ has a Lipschitz bound of constant $L_V dt$:

$$D(V_\theta^\dagger(\mathbf{z})|\alpha\rangle, V_\theta^\dagger(\mathbf{z}')|\alpha\rangle) \leq L_V dt D(|\psi_{\mathbf{z}}\rangle, |\psi_{\mathbf{z}'}\rangle), \quad (\text{A35})$$

for any pure state $|\alpha\rangle$, $\mathbf{z}$ and $\mathbf{z}'$. Then by choosing a stopping time $T = \frac{C}{L_v} \ln(1/\epsilon)$ with $0 < C < 1$, the 1-Wasserstein distance between the learned ensemble p_θ and the true ensemble p_{true} is bounded by

$$W_1(p_\theta, p_{\text{true}}) \leq \frac{\sqrt{2}}{L_v} \epsilon^{1-C} + \epsilon^{-C} E_{\text{diff}}(T) \quad (\text{A36})$$

with $E_{\text{diff}}(T)$ decaying exponentially as $ne^{-\frac{4\gamma}{3n}T}$ with respect to the diffusion time T . If we choose a sufficiently large measurement strength γ such that $\frac{4\gamma}{3nL_v} \geq 1$, the leading-order behavior of the bound for small ϵ is dominated by the first term in Eq. (A36), and $W_1(p_\theta, p_{\text{true}}) \rightarrow 0$ as $\epsilon \rightarrow 0$.

Proof. We first note that once the learned control unitary V_θ is determined, the reversed state $|\phi_t\rangle$ at any time $0 \leq t \leq T$ generated by V_θ backward from a state $|\phi_T\rangle$ at time T is deterministic:

$$|\phi_t\rangle = \mathcal{T}^\dagger \prod_{t'=t}^{T-dt} V_\theta^\dagger(\mathbf{z}_{t'+dt}^\phi, t') |\phi_T\rangle = V^\dagger(\phi_T, t, T) |\phi_T\rangle, \quad (\text{A37})$$

where $\mathcal{T}^\dagger$ stands for the anti-time ordering, and $\mathbf{z}_t^\phi$ is the state vector of state $|\phi_t\rangle$.

We denote the true probability distribution at time t as $p_{\text{true},t}$, which is evolved by the forward weak measurement diffusion from the initial true probability distribution p_{true} . Similarly, we denote the learned probability distribution at time t as $p_{\theta,t}$, which is generated by reverse diffusion V_θ from the true probability distribution $p_{\text{true},T}$ at time T .

Using the fact that $|\phi_t\rangle = V_\theta^\dagger(\mathbf{z}_{t+dt}^\phi, t) |\phi_{t+dt}\rangle$,

$$\begin{aligned} D(|\phi_t\rangle, |\psi_t\rangle) &= D(V_\theta^\dagger(\mathbf{z}_{t+dt}^\phi, t) |\phi_{t+dt}\rangle, |\psi_t\rangle) = D(|\phi_{t+dt}\rangle, V_\theta(\mathbf{z}_{t+dt}^\phi, t) |\psi_t\rangle) \\ &\leq D(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) + D(|\psi_{t+dt}\rangle, V_\theta(\mathbf{z}_{t+dt}^\phi, t) |\psi_t\rangle) \\ &= D(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) + \sqrt{2 \left(1 - \left| \langle \psi_{t+dt} | V_\theta(\mathbf{z}_{t+dt}^\phi, t) | \psi_t \rangle \right|^2 \right)}. \end{aligned} \quad (\text{A38})$$

which holds for any state $|\psi_{t+dt}\rangle$.

Assume the Wasserstein distance $W_1(p_{\theta,t+dt}, p_{\text{true},t+dt})$ for time $t+dt$ is achieved by the coupling $\pi_{t+dt}^{\text{inf}} \in \Pi(p_{\theta,t+dt}, p_{\text{true},t+dt})$, namely,

$$W_1(p_{\theta,t+dt}, p_{\text{true},t+dt}) = \int D(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) \pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) d\phi_{t+dt} d\psi_{t+dt}. \quad (\text{A39})$$

We define an induced coupling at time t as π_t^d :

$$\pi_t^d(|\phi_t\rangle, |\psi_t\rangle) d\phi_t = d\phi_{t+dt} \int_{\psi_{t+dt}} \pi_{t+dt}^{\text{inf}}(\phi_{t+dt}, |\psi_{t+dt}\rangle) \frac{\Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle)}{p_{\text{true},t+dt}(|\psi_{t+dt}\rangle)} d\psi_{t+dt}, \quad (\text{A40})$$

where $|\phi_{t+dt}\rangle = V_\theta(\mathbf{z}_{t+dt}^\phi, t) |\phi_t\rangle$. The function $\Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle)$ is the true joint distribution between states $|\psi_{t+dt}\rangle, |\psi_t\rangle$, satisfying

$$\int \Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle) d\psi_{t+dt} = p_{\text{true},t}(|\psi_t\rangle), \quad \int \Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle) d\psi_t = p_{\text{true},t+dt}(|\psi_{t+dt}\rangle).$$

Note that $\frac{\Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle)}{p_{\text{true},t+dt}(|\psi_{t+dt}\rangle)} = p_{\text{true}}(|\psi_t| | \psi_{t+dt})$ is the conditional probability of $|\psi_t\rangle$ with respect to $|\psi_{t+dt}\rangle$ of the true distribution, which satisfies $\int p_{\text{true}}(|\psi_t| | \psi_{t+dt}) d\psi_t = 1$.

By Eqs. (A38) and (A40), we arrive at the following inequality:

$$\begin{aligned} &\int D(|\phi_t\rangle, |\psi_t\rangle) \pi_t^d(|\phi_t\rangle, |\psi_t\rangle) d\phi_t d\psi_t \\ &\leq \int D(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) \pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) d\phi_{t+dt} d\psi_{t+dt} \\ &\quad + \int \sqrt{2 \left(1 - \left| \langle \psi_{t+dt} | V_\theta(\mathbf{z}_{t+dt}^\phi, t) | \psi_t \rangle \right|^2 \right)} \pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) \frac{\Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle)}{p_{\text{true},t+dt}(|\psi_{t+dt}\rangle)} d\phi_{t+dt} d\psi_{t+dt} d\psi_t \\ &= W_1(p_{\theta,t+dt}, p_{\text{true},t+dt}) + \Delta(\pi_{t+dt}^{\text{inf}}), \end{aligned} \quad (\text{A41})$$

where

$$\Delta(\pi_{t+dt}^{\text{inf}}) = \int \sqrt{2 \left(1 - \left| \langle \psi_{t+dt} | V_\theta(\mathbf{z}_{t+dt}^\phi, t) | \psi_t \rangle \right|^2 \right)} \frac{\pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle)}{p_{\text{true},t+dt}(|\psi_{t+dt}\rangle)} \Gamma_{\text{true},t}(|\psi_{t+dt}\rangle, |\psi_t\rangle) d\phi_{t+dt} d\psi_{t+dt} d\psi_t. \quad (\text{A42})$$

We assume V_θ is Lipschitz with coefficient $L_V dt$, that for any state $|\alpha\rangle$,

$$D(V_\theta^\dagger(\mathbf{z})|\alpha\rangle, V_\theta^\dagger(\mathbf{z}')|\alpha\rangle) \leq L_V dt D(|\psi_{\mathbf{z}}\rangle, |\psi_{\mathbf{z}'}\rangle). \quad (\text{A43})$$

Then, using Eq. (A34), assuming $\mathbf{z}_{t+dt}$ denotes the state vector of state $|\psi_{t+dt}\rangle$, we find

$$\begin{aligned}
\Delta(\pi_{t+dt}^{\text{inf}}) &\leq \sqrt{2}\epsilon dt + \int \left(D(|\psi_t\rangle, V_\theta^\dagger(\mathbf{z}_{t+dt}, t)|\psi_{t+dt}\rangle) - D(|\psi_t\rangle, V_\theta^\dagger(\mathbf{z}_{t+dt}^\phi, t)|\psi_{t+dt}\rangle) \right) \\
&\quad \times \frac{\pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle)}{p_{\text{true}, t+dt}(|\psi_{t+dt}\rangle)} \Gamma_{\text{true}, t}(|\psi_{t+dt}\rangle, |\psi_t\rangle) d\phi_{t+dt} d\psi_{t+dt} d\psi_t \\
&\leq \sqrt{2}\epsilon dt + \int D(V_\theta^\dagger(\mathbf{z}_{t+dt}, t)|\psi_{t+dt}\rangle, V_\theta^\dagger(\mathbf{z}_{t+dt}^\phi, t)|\psi_{t+dt}\rangle) \frac{\pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle)}{p_{\text{true}, t+dt}(|\psi_{t+dt}\rangle)} \Gamma_{\text{true}, t}(|\psi_{t+dt}\rangle, |\psi_t\rangle) d\phi_{t+dt} d\psi_{t+dt} d\psi_t \\
&\leq \sqrt{2}\epsilon dt + L_v dt \int D(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) \pi_{t+dt}^{\text{inf}}(|\phi_{t+dt}\rangle, |\psi_{t+dt}\rangle) d\phi_{t+dt} d\psi_{t+dt} \\
&= \sqrt{2}\epsilon dt + L_v dt W_1(p_{\theta, t+dt}, p_{\text{true}, t+dt}) .
\end{aligned} \tag{A44}$$

Thus, by Eq. (A41) and the definition of the 1-Wasserstein distance, we find

$$W_1(p_{\theta, t}, p_{\text{true}, t}) \leq (1 + L_v dt) W_1(p_{\theta, t+dt}, p_{\text{true}, t+dt}) + \sqrt{2}\epsilon dt . \tag{A45}$$

This implies

$$e^{-L_v(T-t)} W_1(p_{\theta, t}, p_{\text{true}, t}) \leq e^{-L_v(T-t-dt)} W_1(p_{\theta, t+dt}, p_{\text{true}, t+dt}) + \sqrt{2}e^{-L_v(T-t)} \epsilon dt . \tag{A46}$$

At $t = T$, we start to perform the unitary reverse from a uniform random product state distribution $p_{\theta, T}$. Meanwhile, $E_{\text{diff}}(T) \sim n \exp(-\frac{4\gamma}{3n}T)$, because for non-identity Pauli strings, each Pauli weight decays exponentially to zero with increasing diffusion time T , as shown in Eq. (B18). So

$$W_1(p_{\theta, T}, p_{\text{true}, T}) = E_{\text{diff}}(T) \sim n e^{-\frac{4\gamma}{3n}T} , \tag{A47}$$

as is evident from the Pauli weight of the weak measurement channel in Eq. (B18). This together with integrating Eq. (A46) from T to t yields an accumulated bound:

$$W_1(p_{\theta, t}, p_{\text{true}, t}) \leq e^{L_v(T-t)} \left[\frac{\sqrt{2}\epsilon}{L_v} (1 - e^{-L_v(T-t)}) + E_{\text{diff}}(T) \right] . \tag{A48}$$

Thus, particularly, for $t = 0$, the 1-Wasserstein distance between $p_\theta = p_{\theta, 0}$ and $p_{\text{true}} = p_{\text{true}, 0}$ is bounded by

$$W_1(p_\theta, p_{\text{true}}) \leq e^{L_v T} \left[\frac{\sqrt{2}\epsilon}{L_v} (1 - e^{-L_v T}) + E_{\text{diff}}(T) \right] . \tag{A49}$$

Now if we choose a stopping time $T = \frac{C}{L_v} \ln(1/\epsilon)$, then Eq. (A49) becomes

$$W_1(p_0, p_{\text{true}}) \leq \frac{\sqrt{2}}{L_v} \epsilon^{1-C} + \epsilon^{-C} E_{\text{diff}}(T) \tag{A50}$$

In the leading-order behavior of this expression for small ϵ , the second term in Eq. (A50) becomes

$$E_{\text{diff}}(T) \sim n \exp\left(-\frac{4\gamma}{3nL_v} \ln \frac{1}{\epsilon}\right) = n \epsilon^{\frac{4\gamma}{3nL_v}} \tag{A51}$$

In choosing a sufficiently large measurement strength γ , we expect $E_{\text{diff}}(T)$ to decay sufficiently fast that $\frac{4\gamma}{3nL_v} \geq C$, and the leading-order behavior of the bound for small ϵ is then dominated by the first term in Eq. (A50), which is a constant. As $\epsilon \rightarrow 0$,

$$W_1(p_0, p_{\text{true}}) \rightarrow \frac{\sqrt{2}}{L_v} \epsilon^{1-C} \tag{A52}$$

This means that the distribution distance $W_1 \rightarrow 0$ when the training error ϵ is made arbitrarily tiny. $\square$

Appendix B: Pauli Weights Evolution under Measurement Channel and Measurement-and-prepare channel

This appendix derives the analytical expression of the measurement channel $\mathcal{F}_t$ and the measurement-and-prepare channel $\mathcal{M}_t$ we defined in the main text, by solving the differential equation of their Pauli weights.

1. The Measurement channel and the linear SDE

a. The measurement channel

We next examine the generic weak measurement protocol of App. A 2 a in the context of qubit systems. We consider a system of n qubits, initialized in a state ρ_0 . The forward diffusion is implemented by applying a sequence of ancilla-assisted weak measurements. At each infinitesimal time step δt , a single-qubit observable O_t is measured once on a qubit j_t (either randomly chosen or consecutively). The observable is selected from the set of single-qubit Pauli operators $\{\sigma_{x,j}, \sigma_{y,j}, \sigma_{z,j} \mid j = 1, \dots, n\}$ or their linear superpositions, ensuring $O_t^2 = I$. The corresponding Kraus operator is given by:

$$K_{\delta t}(O_t, \delta o_t) = \frac{1}{\sqrt{2}} \exp\left(-\gamma \delta t + 2\gamma O_t \delta o_t\right), \quad \delta o_t = \frac{1}{2} \sqrt{\frac{\delta t}{\gamma}} o_t, \quad o_t = \pm 1, \quad (\text{B1})$$

as given in Eq. (18) of the main text.

In the following derivations of this appendix, for convenience, we do not distinguish between the discrete time δt and dt , and we will denote Eq. (B1) as

$$K_{dt}(O_t, do_t) = \frac{1}{\sqrt{2}} \exp\left(-\gamma dt + o_t \sqrt{\gamma dt} O_{j_t,t}\right) \approx \frac{1}{\sqrt{2}} \left[\left(1 - \frac{\gamma dt}{2}\right) I + o_t \sqrt{\gamma dt} O_t \right]. \quad (\text{B2})$$

Note that $K_{dt}(O_t, do_t)$ differs from Eq. (10) of the main text in normalization. This is because the main text Eq. (10) corresponds to many weak measurement steps within time dt (thus the accumulated signal do_t can be regarded as continuous), while here in Eq. (B2), for simplicity we assume there is only one weak measurement step with discrete outcomes $do_t = \frac{1}{2} \sqrt{\frac{dt}{\gamma}}$.

For a sequence of consecutive weak measurements on the n -qubit system from time $t = 0$ to $t = T$ ($t = kdt$, $1 \leq k \leq N_T$, $N_T = T/dt$), we define a trajectory $\mathcal{O}$ as the sequence of measured observables O_t and their corresponding outcomes $o_t \in \{\pm 1\}$ at each time step:

$$\mathcal{O} = \{(O_t, o_t) \mid 0 \leq t \leq T\} \quad (\text{B3})$$

The associated Kraus operator up to time t along the trajectory $\mathcal{O}$ is

$$K_t(\mathcal{O}) = \mathcal{T} \prod_{t'=0}^t K_{dt}(O_{t'}, do_{t'}), \quad (\text{B4})$$

where $\mathcal{T}$ denotes the time-ordering operator, and K_{dt} is defined in Eq. (B2). At time t , the system state ρ_t conditional on trajectory $\mathcal{O}$ is given by

$$\rho_t(\mathcal{O}) = \frac{\tilde{\rho}_t(\mathcal{O})}{\text{Tr}[\tilde{\rho}_t(\mathcal{O})]}, \quad \tilde{\rho}_t(\mathcal{O}) = 2^{N_t} K_t(\mathcal{O}) \rho_0 K_t^\dagger(\mathcal{O}), \quad N_t = \frac{t}{dt}. \quad (\text{B5})$$

where ρ_0 is the initial state at $t = 0$, and we have defined a tilded state $\tilde{\rho}_t(\mathcal{O})$ with unnormalized trace. The trace of $\tilde{\rho}_t(\mathcal{O})$ has the physical meaning of being the probability for trajectory $\mathcal{O}$ to occur:

$$p_t(\mathcal{O}) = \frac{1}{2^{N_t}} \text{Tr}[\tilde{\rho}_t(\mathcal{O})]. \quad (\text{B6})$$

As defined in the main text, the measurement channel $\mathcal{F}_t$ is the average of the final states of all trajectories respecting their probabilities:

$$\mathcal{F}_t(\rho_0) = \sum_{\mathcal{O}} p_t(\mathcal{O}) \rho_t(\mathcal{O}) = \frac{1}{2^{N_t}} \sum_{\mathcal{O}} \tilde{\rho}_t(\mathcal{O}) = \frac{1}{2^{N_t}} \sum_{\mathcal{O}} K_t(\mathcal{O}) \rho_0 K_t^\dagger(\mathcal{O}). \quad (\text{B7})$$

Particularly, in the final expression, all the 2^{N_t} trajectories $\mathcal{O}$ is summed at equal weight, as their probability $p_t(\mathcal{O})$ is readily included in the trace of $\tilde{\rho}_t(\mathcal{O})$. Therefore, we say that the unnormalized states $\tilde{\rho}_t(\mathcal{O})$ of different trajectories $\mathcal{O}$ effectively occur at equal probability.

b. The linear SDE

The unnormalized state $\tilde{\rho}_t(\mathcal{O})$ defined in Eq. (B5) plays an important role in the above quantum channels, so it is useful to take a closer look at it. Its evolution under the sequence of weak measurements is governed by a linear SDE [31]:

$$d\tilde{\rho}_t = -\frac{\gamma}{2}[O_t, [O_t, \tilde{\rho}_t]]dt + \sqrt{\gamma}\{O_t, \tilde{\rho}_t\}d\tilde{w}_t, \quad d\tilde{w}_t = o_t\sqrt{dt} = 2\sqrt{\gamma}do_t, \quad (\text{B8})$$

which can be derived from the expression of Kraus operators K_{dt} , and $d\tilde{w}_t$ is the stochastic noise variable. Note that for later convenience, we choose to rescale the variable do_t into $d\tilde{w}_t$ according to the above here. An important difference compared to the normalized state ρ_t , since each unnormalized state $\tilde{\rho}_t$ occurs at equal probability (Eq. (B7)), $d\tilde{w}_t$ in the linear SDE Eq. (B8) for $\tilde{\rho}_t$ has a probability distribution with first and second moment

$$\mathbb{E}[d\tilde{w}_t] = 0, \quad \mathbb{E}[d\tilde{w}_t^2] = dt. \quad (\text{B9})$$

In the limit $dt \rightarrow 0$, it is sufficient to keep all the calculations up to order $dt \sim d\tilde{w}_t^2$, so all probability distributions of $d\tilde{w}_t$ with the same first and second moments are equivalent for the SDE. Thus, we can simply set $d\tilde{w}_t$ as the standard random Wiener noise obeying a Gaussian distribution satisfying Eq. (B9).

in terms of $d\tilde{w}_t$, the infinitesimal Kraus operator in Eq. (B2) can be rewritten as

$$K_{dt}(t, d\tilde{w}_t) = \frac{1}{\sqrt{2}} \exp(-\gamma dt + \sqrt{\gamma} O_t d\tilde{w}_t), \quad (\text{B10})$$

which will be used hereafter in quantum channel calculations.

c. Solving the measurement channel

As the equal probability average over all $\tilde{\rho}_t(\mathcal{O})$, the measurement channel satisfies a differential equation being the average of the linear SDE in Eq. (B8), namely, for infinitesimal evolution:

$$d\rho_t = \mathcal{F}_{dt}(\rho_t) - \rho_t = -\frac{\gamma}{2}[O_t, [O_t, \rho_t]]dt. \quad (\text{B11})$$

Assume O_t acts on qubit j_t . By averaging O_t over all possible single-qubit Pauli operators ($\sigma_{j_t, x} = X_j, \sigma_{j_t, y} = Y_j, \sigma_{j_t, z} = Z_j$ each with 1/3 probability) on qubit j_t , we find

$$d\rho_t = \mathcal{F}_{dt}(\rho_t) - \rho_t = -\frac{\gamma}{6} \sum_{\mu=x,y,z} [\sigma_{j_t, \mu}, [\sigma_{j_t, \mu}, \rho_t]]dt. \quad (\text{B12})$$

Particularly, this shows that $\mathcal{F}_{dt}$ maps a Pauli string operator P to itself, and thus the measurement channel is diagonal in the Pauli basis. More explicitly, we can expand the density matrix in the Pauli basis as

$$\rho_t = \frac{1}{2^n} \sum_i z_{t,i} P_i = \frac{1}{2^n} \mathbf{z}_t \cdot \mathbf{P}, \quad \rho_{t+dt} = \mathcal{F}_{dt}(\rho_t), \quad (\text{B13})$$

and the above differential equation implies

$$dz_{t,i} = -\frac{\gamma dt}{6} z_{t,i} \sum_{\mu=x,y,z} [\sigma_{j_t, \mu}, [\sigma_{j_t, \mu}, P_i]] = -\frac{4\gamma dt}{3} z_{t,i} \delta_{|P_i|, j_t}, \quad (\text{B14})$$

where $|P|$ denotes the support of Pauli operator P , while we define $\delta_{|P|, j_t} = 1$ if qubit $j_t \in |P|$, and $\delta_{|P|, j_t} = 0$ otherwise. Solving the above equation gives the following solution:

$$z_{t,i} = w_{\mathcal{F}_t}(P_i) z_{0,i}, \quad w_{\mathcal{F}_t}(P) = \exp\left(-\frac{4\gamma}{3} \sum_{j \in |P|} t^{(j)}\right), \quad (\text{B15})$$

where we have defined $t^{(j)}$ as the total length of time that the j -th qubit is measured (in other words, the j -th qubit is measured for $t^{(j)}/dt$ times), which satisfy $\sum_{j=1}^n t^{(j)} = t$. The coefficient $w_{\mathcal{F}_t}(P)$ is called the *Pauli weight* of the measurement channel, which is defined as

$$w_{\mathcal{F}_t}(P) = \frac{\text{Tr}[P \mathcal{F}_t(P)]}{\text{Tr} \mathbf{1}}, \quad \rightarrow \quad \mathcal{F}_t(P) = w_{\mathcal{F}_t}(P) P. \quad (\text{B16})$$

Thus, the action of the measurement channel on a density matrix $\rho_0 = \frac{1}{2^n} \sum_i z_{0,i} P_i$ is

$$\mathcal{F}_t \left(\frac{1}{2^n} \sum_i z_{0,i} P_i \right) = \frac{1}{2^n} \sum_i w_{\mathcal{F}_t}(P) z_{0,i} P_i . \quad (\text{B17})$$

For weak measurements either uniformly randomly on a qubit, or sequentially on each of the n qubits, in the long time limit $t/dt \gg n$, each qubit is on average measured for a total length of time $t^{(j)} = t/n$. Thus, the Pauli weight in Eq. (B15) becomes

$$w_{\mathcal{F}_t}(P) = w_{\mathcal{F},m}(t) = \exp \left(-\frac{4\gamma}{3n} m t \right) , \quad m = |P| , \quad (\text{B18})$$

where $m = |P|$ denotes the support size of the Pauli string operator P .

2. The Measurement-and-prepare channel

a. Definition of the channel

We now turn to the measurement-and-prepare channel, which is essential for our method to extract weak measurement shadow tomography from the outcomes gathered through weak measurements.

We define the following unnormalized classical snapshot state ($\text{Tr}[\sigma_t(\mathcal{O})] \neq 1$):

$$\sigma_t(\mathcal{O}) = K_t^\dagger(\mathcal{O}) K_t(\mathcal{O}) , \quad (\text{B19})$$

which tends to unnormalized random pure products state when $t \rightarrow \infty$. The random single qubit measurements then form a measurement-and-prepare quantum channel $\mathcal{M}_t$ that maps the initial state ρ_0 into

$$\mathcal{M}_t(\rho_0) = \frac{1}{2^{N_t}} \sum_{\mathcal{O}} \sigma_t(\mathcal{O}) \text{Tr}[\sigma_t(\mathcal{O}) \rho_0] = \frac{1}{2^{N_t}} \sum_{\mathcal{O}} \sigma_t(\mathcal{O}) p_t(\mathcal{O}) = \mathbb{E}_{\mathcal{O}} \sigma_t(\mathcal{O}) p_t(\mathcal{O}) = \mathbb{E}_{\mathcal{O} \sim p(\mathcal{O}|\rho_0)} \sigma_t(\mathcal{O}) , \quad (\text{B20})$$

where $p_t(\mathcal{O})$ is the probability of trajectory $\mathcal{O}$ as defined in Eq. (B6), $N_t = t/dt$ is the number of steps. The channel in Eq. (B20) is defined not preserving the trace of ρ_0 . The randomized measurement channel maps the underlying state ρ to an ensemble of classical snapshot states $\sigma \sim p(\sigma|\rho) \sim \text{Tr}(\sigma\rho)$.

In practice, $\mathcal{M}_t(\rho_0)$ can be measured by summing over the output snapshot states $\sigma_t(\mathcal{O})$ of all the trajectories measured. By definition, each trajectory $\mathcal{O}$ occurs with probability $p_t(\mathcal{O})$, thus the resulting sum approaches $\mathcal{M}_t(\rho_0)$. Note that the output $\sigma_t(\mathcal{O})$ of the measurement-and-prepare channel can be calculated solely from the measurement outcomes which determines $K_t(\mathcal{O})$, and no information about ρ_0 is needed. By reversing the measurement-and-prepare channel, one can deduce ρ_0 , or the shadow of it. Thus, it is important to derive the channel $\mathcal{M}_t$ analytically, which we shall do below.

As local scrambled channels (because of randomized single-qubit measurements invariant under local SU(2) rotations), both channels are diagonal in the Pauli basis, and we can define their Pauli weight as:

$$w_{\mathcal{M}_t}(P) = \frac{\text{Tr}[P \mathcal{M}_t(P)]}{\text{Tr} \mathbf{1}} . \quad (\text{B21})$$

Accordingly, their action on Pauli operator P is given by

$$\mathcal{M}_t(P) = w_{\mathcal{M}_t}(P) P . \quad (\text{B22})$$

Particularly, $\text{Tr}[\mathcal{M}_t(\rho_0)] = w_{\mathcal{M}_t}(\mathbf{1})$ is unnormalized. In the below, we derive the Pauli weight $w_{\mathcal{M}_t}(P)$ of the unnormalized channel.

b. Pauli Transfer Matrix

For simplicity, we ignore the trajectory label $\mathcal{O}$ in this subsection. We want to find a differential equation satisfied by the unnormalized Pauli weight $w_{\mathcal{M}_t}(P)$. To do this, we consider the time evolution of n steps in time ndt , in which

we assume each of the n qubits is measured once. We note that the unnormalized channel in Eq. (B20) at time t and $t + ndt$ can be rewritten as

$$\mathcal{M}_t(\rho_0) = \mathbb{E}_{\sigma_t} \sigma_t \text{Tr}[\sigma_t \rho_0] , \quad \mathcal{M}_{t+ndt}(\rho_0) = \mathbb{E}_{\mathcal{N}} \mathbb{E}_{\sigma_t} \mathcal{N}(\sigma_t) \text{Tr}[\mathcal{N}(\sigma_t) \rho_0] , \quad (\text{B23})$$

where we have defined a channel $\mathcal{N}$ as

$$\sigma_{t+ndt} = \mathcal{N}(\sigma_t) = 2^n K_{ndt}^\dagger \sigma_t K_{ndt} . \quad (\text{B24})$$

Here K_{ndt} is a random Kraus operator corresponding to n steps of weak measurement, which has the form

$$K_{ndt} = \exp \left(-n\gamma dt + \sum_{j=1}^n \sqrt{\gamma} O_j d\tilde{w}_j \right) \simeq 1 + \sum_{j=1}^n \sqrt{\gamma} O_j d\tilde{w}_j - \frac{n\gamma}{2} dt , \quad (\text{B25})$$

with O_j being the random Pauli operator measured on qubit j ($O_j^2 = 1$), and $d\tilde{w}_j = \pm\sqrt{dt}$ are random measurement outcomes.

It is clear that the channel $\mathcal{N}$ above is also locally scrambled, namely, its distribution is invariant under local SU(2) rotations of each qubit. Therefore, by [32], the evolution of Pauli weights from $\mathcal{M}_t$ to $\mathcal{M}_{t+ndt}$ defined above is given by a linear transformation governed by the *Pauli transfer matrix* (PTM):

$$w_{\mathcal{M}_{t+ndt}}(P) = \sum_{P'} W(P, P') w_{\mathcal{M}_t}(P', t) , \quad (\text{B26})$$

where P runs over all the Pauli string operators, and the PTM is defined by

$$W(P, P') = \mathbb{E}_{\mathcal{N}} \left(\frac{\text{Tr}[P \mathcal{N}(P')]}{\text{Tr} \mathbb{1}} \right)^2 . \quad (\text{B27})$$

Calculating the PTM therefore gives the differential equations satisfied by Pauli weights $w_{\mathcal{M}_t}(P)$.

c. Deriving the Pauli weight differential equation

For a Pauli string operator P' with support on $m' = |P'|$ qubits, the action of channel $\mathcal{N}$ to linear order of dt reads:

$$\begin{aligned} \mathcal{N}(P') &\simeq \left(1 + \sum_{j=1}^n \sqrt{\gamma} O_j d\tilde{w}_j - \frac{\gamma}{2} ndt \right) P' \left(1 + \sum_{j=1}^n \sqrt{\gamma} O_j d\tilde{w}_j - \frac{\gamma}{2} ndt \right) \\ &\simeq (1 - \gamma q dt) P' + \sum_{j=1}^n \sqrt{\gamma} d\tilde{w}_j \{O_j, P'\} + \sum_{j,j'} \gamma d\tilde{w}_j d\tilde{w}_{j'} O_j P' O_{j'} , \end{aligned} \quad (\text{B28})$$

from which we see the channel $\mathcal{N}$ can map a Pauli operator P into another Pauli operator P' with the support size (1) increase by 1, or (2) invariant, or (3) decrease by 1. Therefore, we expect the PTM $W(P, P') \neq 0$ only if $|m - m'| \leq 1$, where $m = |P|$ and $|m'| = |P'|$ are the support size of Pauli operators P and P' .

Explicitly, using the definition of PTM in Eq. (B27), up to linear order of dt (note that $d\tilde{w}_j$ is of order $\sqrt{dt}$), and using the fact that $\mathbb{E}[d\tilde{w}_j] = 0$, and $\mathbb{E}[d\tilde{w}_j d\tilde{w}_{j'}] = dt \delta_{jj'}$, we find

$$W_{\mathcal{N}}(P, P') = \left(1 - \frac{8}{3} m \gamma dt \right) \delta_{P, P'} + 4\gamma dt \sum_{j=1}^n \mathbb{E}_{O_j} \left[\delta_{P, \frac{1}{2}\{O_j, P'\}} \right] . \quad (\text{B29})$$

By further acting on $w_{\mathcal{M}_t}(P')$ and averaging over O_j among all single-qubit Pauli operators with equal probability, we arrive at

$$\begin{aligned} w_{\mathcal{M}_{t+ndt}}(P) &= \sum_{P'} W(P, P') w_{\mathcal{M}_t}(P') \\ &= \left(1 - \frac{8}{3} m \gamma dt \right) w_{\mathcal{M}_t}(P) + 4\gamma dt \sum_{P'} \mathbb{E}_{\mathcal{N} \in \mathcal{F}_{\mathcal{N}}} \sum_{j=1}^n \delta_{P, \frac{1}{2}\{O_j, P'\}} w_{\mathcal{M}_t}(P') \\ &= \left(1 - \frac{8}{3} m \gamma dt \right) w_{\mathcal{M}_t}(P) + 4\gamma dt \sum_{P'} w_{\mathcal{M}_t}(P') \left[\frac{1}{3} \delta_{m', m-1} \delta_{P, OP'} + \delta_{m', m+1} \delta_{OP, P'} \right] , \end{aligned} \quad (\text{B30})$$

where we have defined $\delta_{P,OP'} = 1$ if $P = OP'$ for some single-qubit Pauli operator O , and $m = |P|$, $|m'| = |P'|$ are the support sizes.

The sum over P' term accounts for transitions into the state P from other Pauli states, in the bracket of which the first term describes a weight-increasing process, where a support- $(m-1)$ Pauli operator P' becomes P , while the second term describes a weight-decreasing process, where a support- $(m+1)$ Pauli operator P' becomes P .

Dividing the above Eq. (B30) by ndt , we arrive at a differential equation for $w_{\mathcal{M}_t}(P)$ in the continuous-time limit:

$$\frac{dw_{\mathcal{M}_t}(P)}{dt} = -\frac{8m}{3n}\gamma w_{\mathcal{M}_t}(P) + \frac{4\gamma}{n} \sum_{P'} w_{\mathcal{M}_t}(P') \left[\frac{1}{3}\delta_{m',m-1}\delta_{P,OP'} + \delta_{m',m+1}\delta_{OP,P'} \right]. \quad (\text{B31})$$

This differential equation can be further simplified. Due to the permutation symmetry among all qubits and the initial condition symmetric among all qubits, we can assure that the Pauli weight depends only on the support size $m = |P|$ of Pauli operator P :

$$w_{\mathcal{M}_t}(P) = w_{\mathcal{M},m}(t), \quad m = |P|, \quad (\text{B32})$$

and the Pauli weights satisfy a closed system of ODEs from Eq. (B31):

$$\frac{dw_{\mathcal{M},m}}{dt} = -\frac{8m}{3n}\gamma w_{\mathcal{M},m} + \frac{4m}{3n}\gamma w_{\mathcal{M},m-1} + \frac{4(n-m)}{n}\gamma w_{\mathcal{M},m+1}, \quad (0 \leq m \leq n) \quad (\text{B33})$$

This set of ODEs govern the time evolution of the Pauli weights $w_{\mathcal{M},m}(t)$ ($0 \leq m \leq n$) for the weak measurement-and-prepare channel, with boundary conditions $w_{\mathcal{M},-1} = w_{\mathcal{M},n+1} = 0$, and initial conditions

$$w_{\mathcal{M},m}(0) = \delta_{m,0}, \quad (\text{B34})$$

as is evident from the channel definition.

d. Analytical solution of the Pauli weights

The coupled linear ODEs in Eq. (B33) can be compactly expressed in matrix form as

$$\frac{d\mathbf{w}_{\mathcal{M}}}{dt} = D \mathbf{w}_{\mathcal{M}}, \quad \mathbf{w}_{\mathcal{M}} = (w_{\mathcal{M},0}, w_{\mathcal{M},1}, \dots, w_{\mathcal{M},n})^T, \quad (\text{B35})$$

where $\mathbf{w}_{\mathcal{M}}$ is the column vector of Pauli weights across all support sizes $0 \leq m \leq n$. The evolution is governed by the $(n+1) \times (n+1)$ tridiagonal matrix D , the nonzero entries of which are given by:

$$D_{m,m'} = \begin{cases} -\frac{8}{3}\frac{m}{n}\gamma, & \text{if } m' = m \\ \frac{4}{3}\frac{m}{n}\gamma, & \text{if } m' = m-1 \\ \frac{4(n-m)}{n}\gamma, & \text{if } m' = m+1 \\ 0, & \text{otherwise} \end{cases} \quad (\text{B36})$$

The matrix D is non-Hermitian, the eigenvalues and eigenvectors of which are typically complicated to obtain. However, due to its special form, an analytical solution can be obtained by mapping the problem to a well-understood physical system: a quantum spin in a magnetic field. This mapping is realized through a similarity transformation $D' = VDV^{-1}$, which converts D into a Hermitian matrix without changing its eigenvalues.

More generically, we consider a tridiagonal matrix D of the form with nonzero entries $D_{m,m} = c_1m + d_0$, $D_{m,m+1} = c_2(n-m)$, and $D_{m,m-1} = c_0m$. We define a diagonal matrix V with elements

$$V_{m,m'} = \delta_{m,m'} \frac{\left(\sqrt{c_2/c_0}\right)^m}{\sqrt{m!(n-m)!}}.$$

A straightforward calculation of the similarity transformation $D' = VDV^{-1}$ yields the following matrix elements:

$$D'_{m,m'} = (VDV^{-1})_{m,m'} = \begin{cases} \sqrt{c_0 c_2} \sqrt{(m+1)(n-m)}, & \text{if } m' = m+1 \\ \sqrt{c_0 c_2} \sqrt{m(n-m+1)}, & \text{if } m' = m-1 \\ c_1 m + d_0, & \text{if } m' = m \\ 0, & \text{otherwise} \end{cases} \quad (\text{B37})$$

These matrix elements are structurally identical to those of spin operators in a spin- s system, where the total number of qubits is $n = 2s$, and the weight index m is related to the z -component of spin via $m = s + s_z$. In particular, the off-diagonal elements correspond to the matrix elements of the spin raising and lowering operators S_+ and S_- , while the diagonal term matches S_z . As a result, the transformed matrix D' can be written as a linear combination of spin- s operators:

$$D' = VDV^{-1} = (c_1 s + d_0) I + c_1 S_z + 2\sqrt{c_0 c_2} S_x, \quad (\text{B38})$$

which corresponds to the Hamiltonian of a spin- s particle in an effective magnetic field $\mathbf{B} = -(2\sqrt{c_0 c_2}, 0, c_1)$. Particularly, it is known that the eigenvalues of such a Hamiltonian are equally spaced, given by

$$\lambda_p = p\Lambda_1 + (n-p)\Lambda_2, \quad \Lambda_{1,2} = \frac{1}{2} \left(c_1 \pm \sqrt{c_1^2 + 4c_0 c_2} \right). \quad (\text{B39})$$

In our case here, the constants are given by

$$c_0 = \frac{4\gamma}{3n}, \quad c_1 = -\frac{8\gamma}{3n}, \quad c_2 = \frac{4\gamma}{n}, \quad d_0 = 0. \quad (\text{B40})$$

Substituting the coefficients from the table, we obtain the exact p -th eigenvalue λ_p of the evolution matrix D :

$$\lambda_p = \frac{4\gamma}{3} - \frac{4\gamma}{n} p \quad \text{for } p = 0, 1, \dots, n. \quad (\text{B41})$$

The largest eigenvalue corresponds to $p = 0$, giving $\lambda_0 = \frac{4\gamma}{3}$. The general solution for the Pauli weight vector is a linear combination of the corresponding eigenvectors $\mathbf{n}^{(p)}$ of D :

$$\tilde{\mathbf{w}}_{\mathcal{M}}(t) = \sum_{p=0}^n c_p e^{\lambda_p t} \mathbf{n}^{(p)}, \quad (\text{B42})$$

where the coefficients c_p are determined by the initial condition $\mathbf{w}_{\mathcal{M}}(0) = (1, 0, \dots, 0)^T$. After solving for the coefficients, one arrives at the solution:

$$w_{\mathcal{M},m}(t) = w_{\mathcal{M},0}(t) \left(\frac{1 - e^{-\frac{16\gamma t}{3n}}}{3 + e^{-\frac{16\gamma t}{3n}}} \right)^m, \quad w_{\mathcal{M},0}(t) = w_{\mathcal{M},t}(1) = \left(3e^{\frac{4\gamma t}{3n}} + e^{-\frac{4\gamma t}{n}} \right)^n. \quad (\text{B43})$$

We can define the Pauli weight of the channel after normalization as

$$\tilde{w}_{\mathcal{M},m}(t) = \frac{w_{\mathcal{M},m}(t)}{w_{\mathcal{M},0}(t)} = \left(\frac{1 - e^{-\frac{16\gamma t}{3n}}}{3 + e^{-\frac{16\gamma t}{3n}}} \right)^m, \quad \rightarrow \quad \tilde{w}_{\mathcal{M},m}(t) = (\tilde{w}_{\mathcal{M},1}(t))^m. \quad (\text{B44})$$

In the long-time limit $t \rightarrow \infty$, one has

$$\tilde{w}_{\mathcal{M},m}(t \rightarrow \infty) = \frac{w_{\mathcal{M},m}(t \rightarrow \infty)}{w_{\mathcal{M},0}(t \rightarrow \infty)} = \frac{1}{3^m}, \quad (\text{B45})$$

This agrees with the fact that $\sigma_t / \text{Tr}(\sigma_t) = K_t^\dagger K_t / \text{Tr}(K_t^\dagger K_t)$ is a product state. It is a product state of mixed state unless $t \rightarrow \infty$, when it becomes a pure product state.

Fig. 6 shows the Pauli weight $\tilde{w}_{\sigma,m}(t)$ at finite time t calculated analytically from the above (see caption for detailed descriptions).

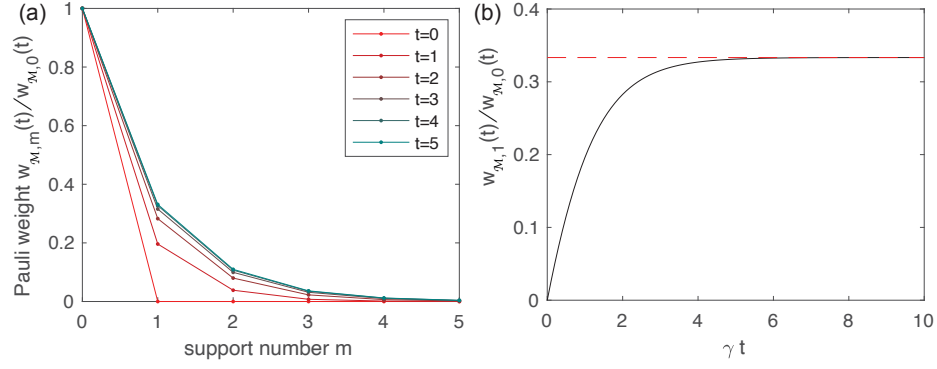


FIG. 6. The total number of qubits is set to $n = 5$, and $\gamma = 1$. (a) Pauli weight $w_{\mathcal{M},m}(t)/w_{\mathcal{M},0}(t)$ as a function of the support size m for different fixed time t (see legend). (b) Pauli weight $w_{\mathcal{M},1}(t)/w_{\mathcal{M},0}(t)$ as a function time t , which approaches $1/3$ as $t \rightarrow \infty$.

3. Error and shadow norm

We recover the initial state ρ_0 by defining shadow of the measured trajectories $\mathcal{O}$:

$$\hat{\rho}_{0,\mathcal{O}} = \mathcal{M}_t^{-1}(\sigma_t(\mathcal{O})) . \quad (\text{B46})$$

It is straightforward to show that the mean value of the shadow of various trajectories $\mathcal{O}$ from the same initial state ρ_0 is the initial state ρ_0 :

$$\mathbb{E}_{p_t(\mathcal{O})} \hat{\rho}_{0,\mathcal{O}} = \mathbb{E}_{\mathcal{O}} p_t(\mathcal{O}) \mathcal{M}_t^{-1}(\sigma_t(\mathcal{O})) = \mathcal{M}_t^{-1} \left[\mathbb{E}_{\mathcal{O}} p_t(\mathcal{O}) \sigma_t(\mathcal{O}) \right] = \mathcal{M}_t^{-1} \left[\mathcal{M}_t(\rho_0) \right] = \rho_0 . \quad (\text{B47})$$

This is the key of our method.

We now examine the variance of an operator O being measured from the shadow. We define

$$\hat{o} = \text{Tr}(\hat{\rho}_{0,\mathcal{O}} O) , \quad \rightarrow \quad \mathbb{E} [\hat{o}] = \text{Tr}(\rho_0 O) . \quad (\text{B48})$$

Its variance is given by

$$\begin{aligned} \text{Var}[\hat{o}] &= \mathbb{E}[(\text{Tr}(\hat{\rho}_{0,\mathcal{O}} O))^2] - (\text{Tr}(\rho_0 O))^2 \leq \mathbb{E}[(\text{Tr}(\hat{\rho}_{0,\mathcal{O}} O))^2] = \mathbb{E}[(\text{Tr}(\mathcal{M}_t^{-1}(O) \sigma_t(\mathcal{O})))^2] \\ &= 2^{N_t} \mathbb{E}_{\mathcal{O}} p_t(\mathcal{O}) (\text{Tr}(\mathcal{M}_t^{-1}(O) \sigma_t(\mathcal{O})))^2 . \end{aligned} \quad (\text{B49})$$

Note that

$$p_t(\mathcal{O}) = \text{Tr}(\sigma_t(\mathcal{O}) \rho_0) \sim \frac{\text{Tr}(\sigma_t(\mathcal{O}))}{\text{Tr} \mathbb{1}} , \quad \sigma_t(\mathcal{O}) = K_t^\dagger(\mathcal{O}) K_t(\mathcal{O}) , \quad (\text{B50})$$

we can estimate the Variance of $\hat{o}$ as

$$\text{Var}[\hat{o}] \leq \mathbb{E}_{\mathcal{O}} \frac{2^{N_t} \text{Tr}(\sigma_t(\mathcal{O}))}{\text{Tr} \mathbb{1}} (\text{Tr}(\mathcal{M}_t^{-1}(O) \sigma_t(\mathcal{O})))^2 = \text{Tr} \left[\mathcal{M}_t^{-1}(O) \mathbb{E}_{\mathcal{O}} \frac{2^{N_t} \text{Tr}(\sigma_t(\mathcal{O}))}{\text{Tr} \mathbb{1}} \sigma_t(\mathcal{O}) \text{Tr}(\mathcal{M}_t^{-1}(O) \sigma_t(\mathcal{O})) \right] . \quad (\text{B51})$$

If we estimate $2^{N_t} \text{Tr}(\sigma_t(\mathcal{O})) \simeq C_t$ as a constant, we approximately have

$$\text{Var}[\hat{o}] \leq \frac{C_t}{\text{Tr} \mathbb{1}} \text{Tr} \left[\mathcal{M}_t^{-1}(O) \mathbb{E}_{\mathcal{O}} \sigma_t(\mathcal{O}) \text{Tr}(\mathcal{M}_t^{-1}(O) \sigma_t(\mathcal{O})) \right] = \frac{C_t}{\text{Tr} \mathbb{1}} \text{Tr} \left[\mathcal{M}_t^{-1}(O) \mathcal{M}_t(\mathcal{M}_t^{-1}(O)) \right] = \frac{C_t}{\text{Tr} \mathbb{1}} \text{Tr} \left[\mathcal{M}_t^{-1}(O) O \right] . \quad (\text{B52})$$

The constant C_t can be fixed by noting that the above upper bound, which is the expectation value of the square of operator O , should be 1 if $O = \mathbb{1}$ is the identity:

$$\frac{C_t}{\text{Tr} \mathbb{1}} \text{Tr} \left[\mathcal{M}_t^{-1}(\mathbb{1}) \right] = 1 , \quad \rightarrow \quad \frac{C_t}{\text{Tr} \mathbb{1}} = \frac{1}{\text{Tr} \left[\mathcal{M}_t^{-1}(\mathbb{1}) \right]} . \quad (\text{B53})$$

This gives the variance of each shadow:

$$\text{Var}[\hat{o}] \leq \|O\|_{\text{sh}}^2, \quad (\text{B54})$$

where we have defined the shadow norm of operator O as

$$\|O\|_{\text{sh}}^2 = \frac{\text{Tr}[\mathcal{M}_t^{-1}(O)O]}{\text{Tr}[\mathcal{M}_t^{-1}(\mathbb{1})]}. \quad (\text{B55})$$

For Pauli operators P with support size $m = |P|$, the shadow norm is

$$\|P\|_{\text{shadow}}^2 = \frac{\text{Tr}[\mathcal{M}_t^{-1}(P)P]}{\text{Tr}\mathbb{1}} = \frac{w_{\mathcal{M}_t}(\mathbb{1})}{w_{\mathcal{M}_t}(P)} \rightarrow 3^m \quad \text{as} \quad t \rightarrow \infty. \quad (\text{B56})$$

For M shadows, the total variance will be $1/M$ factor smaller:

$$\text{Var}[\hat{o}] \leq \frac{\|O\|_{\text{sh}}^2}{M}. \quad (\text{B57})$$

4. Efficient calculation of the shadow

The shadow in Eq. (B46) can be calculated efficiently due to the nice properties we showed earlier. First, recall that the Kraus operator $K_t(\mathcal{O})$ associated with trajectory $\mathcal{O}$ is a product of Kraus operators over each individual qubits:

$$K_t(\mathcal{O}) = \mathcal{T} \prod_{t'=0}^t K_{dt}(O'_t, do'_t) = \otimes_{j=1}^n K_t^{(j)}, \quad K_t^{(j)} = \mathcal{T} \prod_{j_{t'}=j} K_{dt}(O'_t, do'_t), \quad K_{dt}(O_t, do_t) = \frac{1}{\sqrt{2}} \exp\left(-\gamma dt + o_t \sqrt{\gamma dt} O_t\right) \quad (\text{B58})$$

where $\mathcal{T}$ denotes the time-ordering operator, and each Kraus operator $K_t^{(j)}$ is the accumulated contributions of measurements of observables O_t acting on the $j_t = j$ -th qubit. Note that each $K_t^{(j)}$ is simply a 2×2 matrix acting on the j -th qubit.

Therefore, $\sigma_t(\mathcal{O})$ can be reexpressed in a tensor product form:

$$\sigma_t(\mathcal{O}) = \otimes_{j=1}^n \sigma_t^{(j)}(\mathcal{O}), \quad \sigma_t^{(j)}(\mathcal{O}) = K_t^{(j)\dagger}(\mathcal{O}) K_t^{(j)}(\mathcal{O}), \quad (\text{B59})$$

where each $\sigma_t^{(j)}(\mathcal{O})$ is simply a 2×2 matrix.

Then, because the channel $\mathcal{M}_t$ has a Pauli weight $w_{\sigma,m}(t) \equiv (w_{\sigma,1}(t))^m$, the action of $\mathcal{M}_t$ and $\mathcal{M}_t^{-1}$ simply decomposes into the action on each qubit. If we decompose $\sigma_t^{(j)}(\mathcal{O})$ in single-qubit Pauli basis as

$$\sigma_t^{(j)}(\mathcal{O}) = \kappa_0^{(j)} I_j + \kappa_0^{(j)} X_j + \kappa_0^{(j)} Y_j + \kappa_0^{(j)} Z_j, \quad (\text{B60})$$

then we can define the single-qubit shadow (which is a 2×2 matrix)

$$\hat{\rho}_{0,\mathcal{O}}^{(j)} = \mathcal{M}_t^{-1}(\sigma_t^{(j)}) = \kappa_0^{(j)} I_j + \frac{1}{w_{\sigma,1}(t)} \left(\kappa_0^{(j)} X_j + \kappa_0^{(j)} Y_j + \kappa_0^{(j)} Z_j \right), \quad (\text{B61})$$

and the shadow of the total system is simply the tensor product:

$$\hat{\rho}_{0,\mathcal{O}} = \otimes_{j=1}^n \hat{\rho}_{0,\mathcal{O}}^{(j)}. \quad (\text{B62})$$

Appendix C: From measurement channel and reverse Petz channel to quantum diffusion

In this appendix, we show how the weak measurement channel and its Petz recovery channel correspond to the partial differential equations (PDEs) of the quantum diffusion picture of our weak measurements. The key observation is that the density matrix after the quantum channels can be understood as a quantum generalization of the probability distribution in diffusion problem, which satisfies the PDEs of the diffusion.

1. The measurement channel and the forward diffusion

We first focus on the forward evolution of the system by a sequence of weak measurements.

We consider a system of n qubits. As introduced in the main text, the measurement channel $\mathcal{F}_t$ describes the averaged state of all the weak measurement trajectories $\mathcal{O}$. Earlier, we have derived the analytical expression of the measurement channel. For a density matrix $\rho_0 = \frac{1}{2^n} \sum_i z_i P_i$, where P_i runs over all the Pauli string operators

$$\mathcal{F}_t(\rho_0) = \frac{1}{2^n} \sum_i w_{\mathcal{F}_t}(P_i)(t) z_i P_i, \quad w_{\mathcal{F}_t}(P) = w_{\mathcal{F},m}(t) = \exp\left(-\frac{4\gamma}{3n}mt\right), \quad m = |P|. \quad (\text{C1})$$

Here $m = |P|$ is the support size of Pauli operator P . Note that the Pauli weight $w_{\mathcal{F}_t}(P) = w_{\mathcal{F},m}(t)$ only depends on the support size m of operator P .

We consider the measurement channel evolved state $\rho_t = \mathcal{F}_t(\rho_0)$ of some initial state (mixed or pure) ρ_0 . In practice, the random weak measurement along each trajectory $\mathcal{O}$ leads to a random pure product state, since the single-qubit weak measurements eventually collapse each qubit to a single-qubit pure state. This motivates us to define the following pure product coherent state basis:

$$|\mathbf{n}\rangle\langle\mathbf{n}| = \prod_{j=1}^n \left(\frac{I_j + \mathbf{n}_j \cdot \boldsymbol{\sigma}_j}{2} \right), \quad \mathbf{n} = (\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_n), \quad |\mathbf{n}_j| = 1, \quad \mathbf{n}_j = (n_{j,x}, n_{j,y}, n_{j,z}), \quad (\text{C2})$$

where I_j is the identity matrix of the j -th qubit, and $\boldsymbol{\sigma}_j = (\sigma_{j,x}, \sigma_{j,y}, \sigma_{j,z})$ of the j -th qubit. As defined, $\mathbf{n}_j$ is a unit vector living on the Bloch sphere of the j -th qubit ($1 \leq j \leq n$). the coherent states $|\mathbf{n}\rangle$ form an overcomplete basis of the system, satisfying

$$\int d\mathbf{n} |\mathbf{n}\rangle\langle\mathbf{n}| = \frac{I}{2^n}, \quad \int d\mathbf{n} = \text{Tr}\left(\frac{I}{2^n}\right) = 1, \quad (\text{C3})$$

where $d\mathbf{n}$ is the Haar measure in the space of $\mathbf{n}$, and I is the identity operator. Generically, the density matrix ρ_t can be expanded in terms of the coherent state projector basis as:

$$\rho_t = \int d\mathbf{n} p(\mathbf{n}, t) |\mathbf{n}\rangle\langle\mathbf{n}|, \quad \int d\mathbf{n} p(\mathbf{n}, t) = 1, \quad (\text{C4})$$

which is known as the Glauber-Sudarshan P-representation, and $p(\mathbf{n}, t)$ can be called the quasi-probability. For Hermitian operator ρ_t , $p(\mathbf{n}, t)$ is real. We also note that the choice of $p(\mathbf{n}, t)$ is not unique due to the overcompleteness of coherent state basis.

Generically $p(\mathbf{n}, t)$ is not necessarily positive. Our discussion below does not require $p(\mathbf{n}, t)$ to be non-negative; but we will show $p(\mathbf{n}, t)$ is similar to the probability distribution in the classical diffusion problem.

Physical Meaning and relation to previous considerations. If all the $p(\mathbf{n}, t) \geq 0$, then ρ_t has no entanglement between qubits, but only has classical correlations. In this case, ρ_t can be exactly understood as an ensemble of random pure random product states $|\mathbf{n}\rangle\langle\mathbf{n}|$ with probability distribution $p(\mathbf{n}, t)$, which are evolved from an initial state ensemble of random pure product states with probability distribution $p(\mathbf{n}, 0)$. We note that for generic states, we have been denoting the probability distribution of states as $p(\mathbf{z}, t)$, in terms of variable $\mathbf{z}$ which has components z_P being the coefficients of Pauli matrices P in the density matrix $\rho = \frac{1}{2^n} \sum_i z_i P_i$ (see Eq. (C1)). When the initial states of our setup are restricted to random pure product states $|\mathbf{n}\rangle$, they will remain random pure product states during the weak measurement evolution; their corresponding variable $\mathbf{z}$ is a function of $\mathbf{n}$ determined by $|\mathbf{n}\rangle\langle\mathbf{n}| = \frac{1}{2^n} \sum_i z_i P_i$, and by changing of variables we can re-express $p(\mathbf{z}, t)$ as $p(\mathbf{n}, t)$. Accordingly, the nonlinear SDE for variable $\mathbf{z}$ becomes a nonlinear SDE for variable $\mathbf{n}$ on the Bloch spheres.

a. The single-qubit example

We first consider the example of a single-qubit system, for which $\mathbf{n} = \mathbf{n}_1$. In this case, the state $\rho_t = \mathcal{F}_t(\rho_0)$ at time t is given by

$$\rho_t = \int d\mathbf{n}_1 p(\mathbf{n}_1, t) \frac{I_1 + \mathbf{n}_1 \cdot \boldsymbol{\sigma}_1}{2}. \quad (\text{C5})$$

By Eq. (C1), we have $p(\mathbf{n}_1, t)$ satisfying

$$\int d\mathbf{n}_1 p(\mathbf{n}_1, t) = 1, \quad \int d\mathbf{n}_1 p(\mathbf{n}_1, t) \mathbf{n}_1 = e^{-\frac{4\gamma}{3}t} \int d\mathbf{n}_1 p(\mathbf{n}_1, 0) \mathbf{n}_1. \quad (\text{C6})$$

We now show that this correspond to a diffusion problem of quasi-probability distribution $p(\mathbf{n}_1, t)$. As we have explained, ρ_t in the form of Eq. (C5) can be understood as the average of an ensemble of random pure single-qubit states $|\mathbf{n}_1\rangle$ with a “probability” distribution $p(\mathbf{n}_1, t)$. Particularly, ρ_0 corresponds to an ensemble of initial states $|\mathbf{n}_1\rangle$ with a “probability” distribution $p(\mathbf{n}_1, 0)$. In this setup, the nonlinear SDE in terms of variable $\mathbf{z}$, when transformed into variable $\mathbf{n}_1$, gives a diffusion PDE for the “probability” distribution $p(\mathbf{n}_1, t)$ restricted on the Bloch sphere:

$$\partial_t p(\mathbf{n}_1, t) = D \nabla_{\perp}^2 p(\mathbf{n}_1, t), \quad (\text{C7})$$

where $\nabla_{\perp}^2$ is the Laplacian $\nabla_{\mathbf{n}_1}^2$ restricted on the Bloch sphere $|\mathbf{n}_1| = 1$, and D is the diffusion constant which can be derived from the nonlinear SDE restricted on Bloch sphere. Here we shall determine D by matching the evolution of the measurement channel. Note that the PDE in Eq. (C7) has no drifting force, since the diffusion is spherical symmetric on the Bloch sphere. (The drifting force $\mathbf{f}$ in the nonlinear SDE for generic state ρ is solely driving the state towards random pure product states. So for random pure product states, the drifting force is zero.)

We can solve the diffusion equation by expanding $p(\mathbf{n}_1, t)$ in terms of spherical harmonics as

$$p(\mathbf{n}_1, t) = \sum_{lm} \tilde{p}_{lm}(t) Y_{lm}(\theta_1, \varphi_1), \quad \tilde{p}_{lm}(t) = \int d\mathbf{n}_1 p(\mathbf{n}_1, t) Y_{lm}(\theta_1, \varphi_1), \quad \mathbf{n}_1 = (\sin \theta_1 \cos \varphi_1, \sin \theta_1 \sin \varphi_1, \cos \theta_1). \quad (\text{C8})$$

Here (θ_1, φ_1) is the spherical coordinate of $\mathbf{n}_1$. The angular momentum $l = 0, 1, 2, \dots$, and $|m| \leq l$. Eq. (C7) tells us

$$\partial_t \tilde{p}_{lm}(t) = -Dl(l+1) \tilde{p}_{lm}(t), \quad \rightarrow \quad \tilde{p}_{lm}(t) = e^{-Dl(l+1)t} \tilde{p}_{lm}(0). \quad (\text{C9})$$

Note that $Y_{10}(\theta_1, \varphi_1) = \sqrt{\frac{3}{8\pi}} v_{1,z}$, and $Y_{1\pm 1}(\theta_1, \varphi_1) = \sqrt{\frac{3}{8\pi}} (v_{1,x} \pm i v_{1,y})$, from Eqs. (C6) and (C8), we find that $\int d\mathbf{n}_1 p(\mathbf{n}_1, t) \mathbf{n}_1 \propto \tilde{p}_{1m}(t)$, so for $l = 1$,

$$\tilde{p}_{1m}(t) = e^{-\frac{4\gamma}{3}t} \tilde{p}_{1m}(0). \quad (\text{C10})$$

Comparing with Eq. (C9) at $l = 1$, we find the diffusion constant $D = 2\gamma/3$. Thus, ρ_t from the measurement channel is equivalent to the forward diffusion problem of probability distribution $p(\mathbf{n}_1, t)$ satisfying the forward diffusion PDE:

$$\partial_t p(\mathbf{n}_1, t) = \frac{2\gamma}{3} \nabla_{\perp}^2 p(\mathbf{n}_1, t). \quad (\text{C11})$$

Lastly, $\tilde{p}_{00}(t) \equiv \frac{1}{\sqrt{4\pi}}$, which gives the normalized total probability $\int d\mathbf{n}_1 p(\mathbf{n}_1, t) = \sqrt{4\pi} \tilde{p}_{00}(t) = 1$.

b. n -qubits

For n -qubit systems, the generalization is straightforward: the quasi-probability $p(\mathbf{n}, t)$ of ρ_t diffuses simultaneously on the Bloch sphere of every qubit. Accordingly, it satisfy a forward diffusion equation

$$\partial_t p(\mathbf{n}, t) = \frac{2\gamma}{3n} \nabla_{\perp}^2 p(\mathbf{n}, t), \quad \nabla_{\perp}^2 = \nabla_{\perp,1}^2 + \nabla_{\perp,2}^2 + \dots + \nabla_{\perp,n}^2, \quad (\text{C12})$$

where $\nabla_{\perp,j}^2$ is the Laplacian with respect to $\mathbf{n}_j$ restricted on the Bloch sphere $|\mathbf{n}_j| = 1$ of the j -th qubit. Namely, $\nabla_{\perp}^2$ is the Laplacian on the manifold of $\mathbf{n} \in \otimes_{j=1}^n S_j^2$, where S_j^2 is the Bloch sphere of the j -th qubit. It is straightforward to verify that such a diffusion equation for the quasi-probability $p(\mathbf{n}, t)$ is equivalent to the state ρ_t evolved by the measurement channel as given in Eq. (C1).

Physically, this is equivalent to setting the initial state ensemble to be the random pure product states $|\mathbf{n}\rangle\langle\mathbf{n}|$ with a probability distribution $p(\mathbf{n}, 0)$, and evolve every state by a sequence of random weak measurements, where the resulting probability will be $p(\mathbf{n}, t)$ given by Eq. (C12). Here $p(\mathbf{n}, t)$ is related to the probability distribution $p(\mathbf{z}, t)$ we studied earlier for generic states by the variable transformation $|\mathbf{n}\rangle\langle\mathbf{n}| = \frac{1}{2^n} \sum_i z_i P_i$ from $\mathbf{z}$ to $\mathbf{n}$. Accordingly, the SDE for $\mathbf{z}$ transforms into a SDE for $\mathbf{n}$, and the PDE for $p(\mathbf{z}, t)$ transforms to the PDE in Eq. (C12) for $p(\mathbf{n}, t)$.

2. The Petz recovery channel and the backward diffusion

We now consider the Petz recovery channel of the measurement channel. For states $\rho_t = \mathcal{F}_t(\rho_0)$ evolved by the measurement channel $\mathcal{F}_t$ defined above, the Petz recovery channel reversing from final time T to time $T - t$ is defined as

$$\mathcal{R}_t(\sigma) = \rho_{T-t}^{1/2} \mathcal{F}_t^\dagger(\rho_T^{-1/2} \sigma \rho_T^{-1/2}) \rho_{T-t}^{1/2}, \quad \rightarrow \quad \mathcal{R}_t(\rho_T) = \rho_{T-t}, \quad (\text{C13})$$

where $\mathcal{F}_t^\dagger$ is the conjugate of the measurement channel $\mathcal{F}_t$, defined by $\text{Tr}(\mathcal{F}_t(\rho_1)\rho_2) = \text{Tr}(\rho_1\mathcal{F}_t^\dagger(\rho_2))$. In the case of measurement channel in Eq. (C1), it is easy to see that $\mathcal{F}_t^\dagger = \mathcal{F}_t$.

We consider the infinitesimal Petz recovery channel from ρ_{t+dt} to ρ_t given by

$$\mathcal{R}_{dt}(\sigma) = \rho_t^{1/2} \mathcal{F}_{dt}^\dagger(\rho_{t+dt}^{-1/2} \sigma \rho_{t+dt}^{-1/2}) \rho_t^{1/2}, \quad (\text{C14})$$

and show it is analogous to the backward diffusion in the diffusion model. To the linear order of dt , we have

$$\frac{d\sigma}{dt} = \frac{\mathcal{R}_{dt}(\sigma) - \sigma}{dt} = \rho_t^{1/2} \frac{\mathcal{F}_{dt} - 1}{dt} (\rho_t^{-1/2} \sigma \rho_t^{-1/2}) \rho_t^{1/2} + \frac{\rho_t^{1/2} \rho_{t+dt}^{-1/2} - 1}{dt} \sigma + \sigma \frac{\rho_{t+dt}^{-1/2} \rho_t^{1/2} - 1}{dt}. \quad (\text{C15})$$

We rewrite the operators involved above in the Glauber-Sudarshan P-representation as:

$$\rho_t = \int d\mathbf{n} p(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad \rho_t^{1/2} = \int d\mathbf{n} \eta(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad \rho_t^{-1/2} = \int d\mathbf{n} \lambda(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad (\text{C16})$$

and

$$\frac{\rho_t^{1/2} \rho_{t+dt}^{-1/2} - 1}{dt} = \int d\mathbf{n} \beta(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|. \quad (\text{C17})$$

To better understand them, we first define the multiplication of operators in the Glauber-Sudarshan P-representation:

$$A = \int d\mathbf{n} p_A(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad B = \int d\mathbf{n} p_B(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad AB = \int d\mathbf{n} p_{AB}(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad (\text{C18})$$

where we can define a multiplication operation

$$p_{AB}(\mathbf{n}, t) = (p_A * p_B)(\mathbf{n}, t), \quad (\text{C19})$$

where $*$ being an operation similar to convolution. In this sense, the functions defined in Eq. (C16) satisfy $\eta * \eta = p$, and $\lambda * \lambda * p = 1$.

By expressing state σ as

$$\sigma = \rho_t = \int d\mathbf{n} q(\mathbf{n}, t) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad (\text{C20})$$

in terms of a function $q(\mathbf{n}, t)$, we can formally rewrite Eq. (C15) in the Glauber-Sudarshan P-representation as a PDE:

$$\partial_t q = \frac{2\gamma}{3n} \eta * [\nabla_\perp^2 (\lambda * q * \lambda)] * \eta + \beta * q + q * \bar{\beta}, \quad (\text{C21})$$

where $\bar{\beta}$ is the complex conjugate of β , and we have used the fact that the forward evolution $\frac{\mathcal{F}_{dt}-1}{dt} = \frac{2\gamma}{3n} \nabla_\perp^2$ from Eq. (C12).

We claim that Eq. (C21) is the quantum analog of the backward diffusion PDE of the classical diffusion model. To see this, we show it reduces to the classical backward diffusion in a properly defined classical limit.

We define the classical limit as the case where all the coherent states $|\mathbf{n}\rangle$ are orthonormal, namely,

$$\langle \mathbf{n} | \mathbf{n}' \rangle = \delta(\mathbf{n} - \mathbf{n}'). \quad (\text{C22})$$

For qubits, this is an unphysical limit, but we can understand it as being approached when we consider many copies of the coherent states, namely, $|\mathbf{n}\rangle \rightarrow |\mathbf{n}\rangle^{\otimes k}$ with $k \rightarrow \infty$, or equivalently, the large spin limit of each qubit (which is spin 1/2). In this classical limit, the states in the Glauber-Sudarshan P-representation become effectively diagonal,

and by the definition in Eq. (C18), the $*$ multiplication simply becomes the normal multiplication. The functions we defined then reduces to the following (the variable $(\mathbf{n}, t)$ are omitted for simplicity):

$$\eta = p^{1/2}, \quad \lambda = p^{-1/2}, \quad \beta = \frac{p^{1/2}(\mathbf{n}, t)p^{-1/2}(\mathbf{n}, t + dt) - 1}{dt} = -\frac{\partial_t p}{2p} = -\frac{\gamma}{3n} \frac{\nabla_\perp^2 p}{p}. \quad (\text{C23})$$

Substituting these function into Eq. (C21), we find

$$\partial_t q = \frac{2\gamma}{3n} \left[p \nabla_\perp^2 \left(\frac{q}{p} \right) - q \frac{\nabla_\perp^2 p}{p} \right] = \frac{2\gamma}{3n} \left[\nabla_\perp^2 q - 2 \nabla_\perp \cdot \left(q \frac{\nabla_\perp p}{p} \right) \right], \quad (\text{C24})$$

or in a more familiar form,

$$\partial_t q(\mathbf{n}, t) = \frac{2\gamma}{3n} \nabla_\perp^2 q(\mathbf{n}, t) - \frac{4\gamma}{3n} \nabla_\perp \cdot [q(\mathbf{n}, t) \nabla_\perp \log p(\mathbf{n}, t)]. \quad (\text{C25})$$

This is exactly the backward diffusion PDE corresponding to the forward diffusion PDE in Eq. (C12), where $\frac{4\gamma}{3n} \nabla_\perp \log p(\mathbf{n}, t)$ plays the role of the backward drifting force.

Approaching the classical limit. As we have argued above, if we replace each spin 1/2 qubit by a large spin S qudit (which has $2S + 1$ states), Eq. (C22) will be asymptotically satisfied, and we can achieve the classical limit. In this case, state $|\mathbf{n}\rangle$ denotes the pure product of large spin coherent states along direction $\mathbf{n}_j$ on the j -th spin. One then prepares an ensemble of initial states being pure product coherent states $|\mathbf{n}\rangle$ with a probability distribution $p(\mathbf{n}, t)$, and perform a sequence of weak measurement for these large spin qudits. In the large spin limit, the measurement channel $\mathcal{F}_t$ matches exactly with the forward classical diffusion governed by Eq. (C12), and the Petz recovery channel $\mathcal{R}_t$ matches exactly with the backward classical diffusion governed by Eq. (C25).

Appendix D: Lindbladian for twirled Petz map

In this section, we give the expression of the Lindbladian corresponding to the twirled Petz recovery channel.

1. Lindbladian of the measurement channel

For an single dt step of the measurement channel acting on the j -th qubit, the Lindbladian can be derived to be:

$$\mathcal{F}_{dt}(\rho_t) = e^{\mathcal{L}dt}(\rho_t) = \sum_{o_t=\pm 1} K_{dt}(O_t, do_t) \rho_t K_{dt}^\dagger(O_t, do_t), \quad (\text{D1})$$

which gives the Lindbladian equation

$$\frac{d\rho_t}{dt} = \mathcal{L}[\rho_t] = \sum_{\mu=x,y,z} \left(L_\mu \rho_t L_\mu^\dagger - \frac{1}{2} \{L_\mu^\dagger L_\mu, \rho_t\} \right), \quad L_\mu = L_\mu^\dagger = \sqrt{\frac{\gamma}{6}} \sigma_{j,\mu}. \quad (\text{D2})$$

Here the 3 jump operators L_μ runs over all the single-qubit Pauli operators $\sigma_{j,\mu}$ on qubits j . Note that there is no Hamiltonian in the Lindbladian of the measurement channel. This can be obtained by averaging over the linear SDE in the below over all random single qubit operators O_t and averaging over $d\tilde{w}_t$:

$$d\tilde{\rho}_t = -\frac{\gamma}{2} [O_t, [O_t, \tilde{\rho}_t]] dt + \sqrt{\gamma} \{O_t, \tilde{\rho}_t\} d\tilde{w}_t. \quad (\text{D3})$$

Since $L_\mu = L_\mu^\dagger$, we conclude that the conjugate channel $\mathcal{F}_{dt} = \mathcal{F}_{dt}^\dagger$, and the conjugate Lindbladian $\mathcal{L}^\dagger = \mathcal{L}$.

2. Lindbladian of the twirled Petz recovery channel

The generic infinitesimal twirled Petz recovery channel from a generic ρ_{t+dt} to ρ_t is defined as

$$\tilde{\mathcal{R}}_{dt}(\sigma) = \int_{-\infty}^{\infty} f(\tau) \mathcal{R}_{dt}^\tau(\sigma) d\tau, \quad \mathcal{R}_{dt}^\tau(\sigma) = \rho_t^{\frac{1-i\tau}{2}} \mathcal{F}_{dt}^\dagger \left(\rho_{t+dt}^{\frac{-1+i\tau}{2}} \sigma \rho_{t+dt}^{\frac{-1-i\tau}{2}} \right) \rho_t^{\frac{1+i\tau}{2}}, \quad (\text{D4})$$

where $f(\tau) = \frac{1}{2[\cosh(\pi\tau)+1]}$. Setting $\tau = 0$ gives back the untwirled Petz recovery channel. According to Ref. [?], the infinitesimal rotated Petz map $\mathcal{R}_{dt}^\tau(\sigma)$ can be written as a Lindbladian form $\mathcal{R}_{dt}^\tau(\sigma) = e^{\mathcal{L}_{B,\tau}dt}(\sigma)$, with the Lindbladian

$$\mathcal{L}_{B,\tau}(\sigma) = -i[H_{B,\tau}, \sigma] + \sum_{\mu} \left(L_{B,\tau,\mu} \sigma L_{B,\tau,\mu}^\dagger - \frac{1}{2} \{L_{B,\tau,\mu}^\dagger L_{B,\tau,\mu}, \sigma\} \right), \quad (\text{D5})$$

where

$$L_{B,\tau,\mu} = \rho_t^{\frac{1-i\tau}{2}} L_{\mu}^\dagger \rho_t^{\frac{-1+i\tau}{2}}, \quad H_{B,\tau} = -\frac{i}{2} \left[\frac{d(\rho_t^{\frac{1-i\tau}{2}})}{dt} \rho_t^{\frac{-1+i\tau}{2}} + \frac{1}{2} \sum_{\mu} \rho_t^{\frac{1-i\tau}{2}} L_{\mu}^\dagger L_{\mu} \rho_t^{\frac{-1+i\tau}{2}} \right] + h.c. . \quad (\text{D6})$$

Particularly, for our measurement channel, the expression of $H_{B,\tau}$ can be further simplified by noting that $L_{\mu}^\dagger L_{\mu} = \frac{\gamma}{6}$ here is a constant. The Lindbladian for the twirled Petz channel is then given by

$$\tilde{\mathcal{L}}_B(\sigma) = \int_{-\infty}^{\infty} f(\tau) \mathcal{L}_{B,\tau}(\sigma) d\tau, \quad \tilde{\mathcal{R}}_{dt}(\sigma) = e^{\tilde{\mathcal{L}}_B dt}(\sigma). \quad (\text{D7})$$

For our purpose here, we apply the twirled Petz recovery channel $\tilde{\mathcal{R}}_{dt}$ with ρ_{t+dt} and ρ_t in Eq. (D4) replaced by the local reduced density matrix $\rho_t^{S_j}$ to $\rho_{t-dt}^{S_j}$ in subregion S_j around the j -th qubit (defined below). The resulting local twirled Petz recover channel is denoted as $\tilde{\mathcal{R}}_{dt}^{S_j}$, which acts locally on the subregion S_j .

3. Subsystem-Based Density Matrix Estimation

We consider a subsystem S_j centered around qubit j , chosen to include the region that qubit j is entangled with (which is denoted as region B). Denoting qubit j as A , we define $S_j = A \cup B$, with the remainder of the system denoted as C . We impose the condition

$$I(A : C|B) \simeq 0, \quad (\text{D8})$$

which ensures that A is not entangled with C .

To characterize the reduced forward dynamics, we compute the reduced density matrix (RDM) on each subsystem S_j via the following procedure:

1. For each S_j , compute the initial reduced density matrix:

$$\rho_0^{S_j} = \frac{1}{2^{|S_j|}} \sum_{P \in S_j} z_{P,0} P. \quad (\text{D9})$$

2. Verify whether $\rho_0^{S_j}$ is semi-positive definite (SPD). If it is not, deform $\rho_0^{S_j}$ into a valid SPD matrix. This step modifies the Pauli coefficients:

$$z_{P,0} \longrightarrow z_{P,0}^{(S_j)}. \quad (\text{D10})$$

3. Evolve the RDM under the decoherence model:

$$\rho_t^{S_j} = \frac{1}{2^{|S_j|}} \sum_{P \in S_j} z_{P,t}^{(S_j)} P, \quad \text{with} \quad z_{P,t}^{(S_j)} = w_{\mathcal{F}_t}(P) z_{P,0}^{(S_j)} \quad \text{for } P \in S_j. \quad (\text{D11})$$

Similarly, the RDM at time $t - dt$ is given by:

$$\rho_{t-dt}^{S_j} = \frac{1}{2^{|S_j|}} \sum_{P \in S_j} z_{P,t-dt}^{(S_j)} P. \quad (\text{D12})$$

By construction, the SPD property of $\rho_0^{S_j}$ ensures the SPD of $\rho_t^{S_j}$. We assume that $\rho_t^{S_j}$ is a matrix of size $N_{S_j} \times N_{S_j}$, where $N_{S_j} = 2^{|S_j|}$.

Using the approximate data processing inequality [29], there exists a map channel $\tilde{\mathcal{R}}_{dt}^{S_j}$ acting on $S_j = A \cup B$ whose recovery quality satisfies:

$$\frac{1}{2 \ln 2} \left\| \tilde{\mathcal{R}}_{dt}^{S_j} \circ \mathcal{F}_{dt}[\rho_t] - \rho_t \right\|_1^2 \leq I_{\rho_t}(A : C|B) - I_{\mathcal{F}_{dt}[\rho_t]}(A : C|B). \quad (\text{D13})$$

The explicit form of $\tilde{\mathcal{R}}_{dt}^{S_j}$ is given by the twirled Petz recovery map in Eq. (D4), which depends on the forward channel $\mathcal{F}_{dt}$ and the local reduced density matrix $\rho_t^{S_j}$. The above bound shows that if ρ has ξ -FML, local recovery is possible because the recovery error $\left\| \tilde{\mathcal{R}}_{dt}^{S_j} \circ \mathcal{F}_{dt}[\rho_t] - \rho_t \right\|_1$ decays exponentially with the width of B .

Theorem 4. *Let ρ_0 be an initial n -qubit state. The initial state ρ_0 evolves forward under the infinitesimal measurement channel $\mathcal{F}_{dt}$ for total time $T = N_T dt$, resulting in a fully decohered state $\rho_T = \mathcal{F}_T(\rho_0)$. Starting from ρ_T , let the reverse diffusion process be approximated by $N_T = T/dt$ sequential applications of a local twirled Petz recovery channel $\tilde{\mathcal{R}}_{dt}^{S_j}$, defined on local subsystems $S_j = A \cup B$ (that together cover the system). Assume a locality condition such that for every intermediate state ρ_t ($0 \leq t \leq T$) and every choice of subsystem S_j , the conditional mutual information is bounded by $I_{\rho_t}^{S_j}(A : C|B) \leq \delta$, where C is the complement of S_j . Then the final state ρ'_0 produced by applying all N_T Petz recovery steps remains close to the initial state ρ_0 , with the error bounded by the trace distance:*

$$\|\rho'_0 - \rho_0\|_1^2 \leq \frac{2 \ln(2) \cdot \mathcal{O}(n)}{\gamma \cdot dt} \cdot \delta + E_{\text{diff}}(T), \quad (\text{D14})$$

with $E_{\text{diff}}(T)$ decaying exponentially as the diffusion time T increases.

Proof. The total error can be bounded by the sum of single-step errors:

$$\|\rho'_0 - \rho_0\|_1 \leq \sum_{k=1}^{N_T} \left\| \tilde{\mathcal{R}}_{dt}^{S_k}(\mathcal{F}_{dt}(\rho_{t_{k-1}})) - \rho_{t_k} \right\|_1, \quad (\text{D15})$$

where $\rho_{t_k} = \mathcal{F}_{dt}(\rho_{t_{k-1}})$ is the state after the k th forward step of duration dt . By the Cauchy-Schwarz inequality, this implies

$$\|\rho'_0 - \rho_0\|_1^2 \leq N_T \sum_{k=1}^{N_T} \left\| \tilde{\mathcal{R}}_{dt}^{S_k}(\mathcal{F}_{dt}(\rho_{t_{k-1}})) - \rho_{t_k} \right\|_1^2. \quad (\text{D16})$$

Each single-step recovery error is controlled by the drop in the local CMI over that step:

$$\left\| \tilde{\mathcal{R}}_{dt}^S(\mathcal{F}_{dt}(\rho)) - \rho \right\|_1^2 \leq 2 \ln 2 \left[I_\rho(A : C|B) - I_{\mathcal{F}_{dt}(\rho)}(A : C|B) \right], \quad (\text{D17})$$

for any subsystem $S = A \cup B$ (with complement S^C).

Summing these squared errors over $k = 1, \dots, N_T$ yields a telescoping sum of local CMI differences at each recovery step. By the data-processing inequality, the conditional mutual information is non-increasing under the forward channel, $I_{\rho_{t_k}}(A : C|B) \leq I_{\rho_{t_{k-1}}}(A : C|B)$, so the sum of CMI differences telescopes to the initial-minus-final value:

$$\sum_{k=1}^{N_T} \left(I_{\rho_{t_{k-1}}}(A : C|B) - I_{\rho_{t_k}}(A : C|B) \right) = I_{\rho_0}(A : C|B) - I_{\rho_T}(A : C|B) \leq \delta, \quad (\text{D18})$$

using the assumed locality bound on I_{ρ_0} and the fact that $I_{\rho_T} \geq 0$. Combining these results and noting $N_T = T/dt$ gives the stated error bound.

Meanwhile, $E_{\text{diff}}(T) \sim n \exp(-\frac{4\gamma}{3n}T)$, because for non-identity Pauli strings, each Pauli weight decays exponentially to zero with increasing diffusion time T , as shown in Eq. (B18). $\square$